

Rebound effect, emergence and steering in a physical model of a credit society

Anatole Joseph-Stouls

LIED — Laboratoire interdisciplinaire des énergies de demain

(Interdisciplinary Laboratory for Tomorrow's Energies)

anatole.joseph-stouls@ens-rennes.fr

Work in progress — draft translation. This is an English translation of a French working paper still under active revision. It is shared for early feedback and discussion only, and should not be cited or treated as a finished result. Figure axis labels and legends have been regenerated in English throughout; one legacy appendix figure (the stock-versus-flow illustration, figure 25) still carries its original French labels pending regeneration, and its mapping is given in the glossary (section A) and in the body text. The authoritative, up-to-date version is the French original.

Abstract

A local technical improvement does not necessarily translate into a proportional response of the whole system. This work explores the hypothesis that part of the rebound effect might stem from the organization of exchanges, and not only from a change in behavior. We build a physical toy model: entities extract a resource, lend each other productive stock and can disappear, transmitting their losses to their creditors. Population, network and inequality form during the simulation; they are not fixed as targets.

The work articulates three questions: which regularities emerge, which parameters allow them to be varied, and how the system responds to an improvement in its technology. The simulations produce cascades of varied sizes and long-tailed distributions of balance and interest. Power-law fits with cutoff for cascade sizes leave systematic deviations; self-organized criticality is not established. Contractions of activity are short, with a decay of durations close to exponential. The intensity of exchanges acts on propagation and tail statistics; changing the time step does not reduce to a change of unit.

In the main configuration, increasing the extraction coefficient by 50 % increases aggregate production by about 35 %, not 50 %: individual production increases while population decreases. The response depends notably on the size of new entities relative to the productive scale. These results establish an endogenous non-proportionality in the model, not a calibrated measure of the real energy rebound.

1 Introduction: the internship's physical question

The rebound effect denotes the fact that an improvement in energy efficiency does not produce a proportional decrease in consumption. A usual explanation is behavioral: the service becoming cheaper, one consumes more of it. We explore here a different hypothesis, suggested

by work on urban metabolism [1] and by the application of self-organized criticality to the economics of production [2]: *what if part of the response were endogenous to the organization of exchanges*, that is a property of the way agents are coupled rather than a sum of individual decisions?

This question calls for a model in which individual capacities can be modified during the simulation and the ensuing reorganization observed, without a rule having been written to produce a rebound. We build a toy model to this end. It has no prices, no labor, no final consumption, no households distinct from firms: it codes local interactions and lets macroscopic regularities emerge.

Two objectives guide its construction: obtaining a collective response that is not proportional to a microscopic change, and letting classes of distributions emerge that can be confronted, with the necessary precautions, with those observed in the economy. The guiding hypothesis is that an organization exhibiting tendencies toward criticality might amplify or dampen a local perturbation. In a self-organized critical system, activity would spontaneously maintain itself near a propagation threshold; a wide range of event sizes would be an indication of this, not a proof. We do not assume this hypothesis to hold for the model built here.

The stake of the internship is thus not only the last productivity experiment. It includes building a coherent model, searching for relevant endogenous variables, mapping sensitivities, and then running controlled interventions. The reader first meets a four-entity example, before the equations (§2). The protocol distinguishes families of experiments (§3). We then examine emergence (§4), steering and scales (§5), and then the technological response and its distribution (§6). Secondary diagnostics and their limitations are developed in the appendix.

2 A minimal physical and accounting model

2.1 Building intuition with four entities

Imagine four entities able to extract a resource from their environment. Each has a mobilizable stock K_i : the larger it is, the more it extracts, but each additional unit of stock yields less than the previous one. The function $f_i(K) = A_i K^{\gamma_i}$ describes this *extraction technology*. The input joins the stock, a fraction of which decays at each step. New entities appear with the same

endowment K° and, in the reference regime, the same production exponent $\gamma^\circ = 1/2$.

A well-endowed entity can lend stock to another, where that stock produces more at the margin. This real transfer simultaneously creates a claim for the lender and a debt for the borrower. The *principal* q is the nominal amount recorded; the *interest service* rq is the payment due at each step. Interest moves the resource between entities, without creating any. The principal is not progressively repaid in this model: this is a strong institutional assumption.

The loan thus preserves the instantaneous net worth of each partner, but changes its productive capacity and its future commitments. An entity may later be unable to honor its service, or owe more than it holds. Its disappearance wipes out the claims held on it: its lenders can in turn become insolvent. The coupling between real stocks, persistent commitments and propagated losses is the mechanism studied here.

2.2 State, accounting, invariants

The population is open and comprises a single type of agent, called an *entity*. No role – lender, borrower, bank – is declared; all entities follow the same rules, and any observed differentiation is emergent.

The variables K_i, C_i, D_i, NW_i , age and flows received or paid are specific to entity i . The parameters $\lambda, \delta, \sigma, \rho$ and the credit institution are common to the whole population.¹ The technology (A_i, γ_i) is attached to each entity; it is identical for all at the reference point, but can differ after a targeted intervention.

An entity i holds a real productive stock $K_i \geq 0$, counted in joules, which we call its *capital*. This is its intrinsic productive size, not an accounting capital. A loan contract ties a lender to a borrower through a nominal *principal* $q > 0$ and a per-step *rate* $r > 0$. Summing over the contracts in which i is involved gives its claims C_i and its debts D_i , then its *net worth*

$$NW_i = K_i + C_i - D_i. \quad (1)$$

The asymmetry between the two stocks is constitutive: capital is real, productive and depreciable, whereas the principal is nominal, perpetual and not amortized. The debt network can therefore remain heavy while real assets erode.

The rules algebraically enforce, at every step, $K_i \geq 0$, $\sum_i C_i = \sum_i D_i$, the equality between the sum of the net worths of the survivors and their total capital, and the real balance sheet

$$\begin{aligned} \sum_i K_i(t+1) - \sum_i K_i(t) = & \text{endowments} + \text{shock} + \text{production} \\ & - \text{depreciation} - \text{destruction}, \end{aligned} \quad (2)$$

the five terms on the right being recorded separately at each step. The numerical check accounts for floating-point rounding. Loans and interest are purely internal

transfers: there is neither creation nor leakage of joules outside the channels named in (2).

2.3 A time step

The phases run in this order, which is part of the model. Scheduled interventions are applied before births.

- (i) **Births.** A number $B_t \sim \text{Poisson}(\lambda)$ of entities enters, each endowed with a capital K° . The parameter λ is the *birth intensity* and K° the *birth endowment*.
- (ii) **Shock.** $K_i \leftarrow K_i e^{\eta_i}$, with $\eta_i \sim \mathcal{N}(-\sigma^2/2, \sigma^2)$ independent. The multiplicative mean equals one: no drift is imposed. The parameter σ fixes the common amplitude of the noise; the draw η_i is individual and renewed at each step. This is not a shared global shock.
- (iii) **Production.** $K_i \leftarrow K_i + A K_i^\gamma$, with $0 < \gamma < 1$. The parameter A is the *productivity* of the technology and γ its *production exponent*. Concavity is essential: it is what creates a gain from pooling (§2.6).
- (iv) **Interest service.** This is the periodic payment of the loan charge, without repayment of the principal. Each borrower pays $\sum rq$ drawn from its capital; if it cannot, it pays pro rata and enters service default.
- (v) **Depreciation.** $K_i \leftarrow (1-\delta)K_i$, where δ is the *per-step depreciation rate*. It represents the temporal erosion of the stock; its value therefore depends on the duration chosen for a step (§5.4). Principals, for their part, do not depreciate.
- (vi) **Market.** See §2.6.
- (vii) **Bankruptcies.** See §2.7.
- (viii) **Measurements.** No logging option consumes a random draw: trajectories with and without measurement are identical.

2.4 Stocks, flows and the physics–economics correspondence

The parallel between physics and economics is the center of the approach, not an analogy added after the fact. It is nonetheless necessary to distinguish a correspondence of dimensions, a representational hypothesis and an empirical validation. The model is a fundamental study of interactions: the entity remains generic, and can shed light on a concrete system if the relevant mechanisms are indeed present there, without implicitly becoming a person and then a firm.

¹The parameters λ, δ and σ are made explicit in §2.3; ρ and the credit institution in §2.6. The glossary in appendix A gathers the notations.

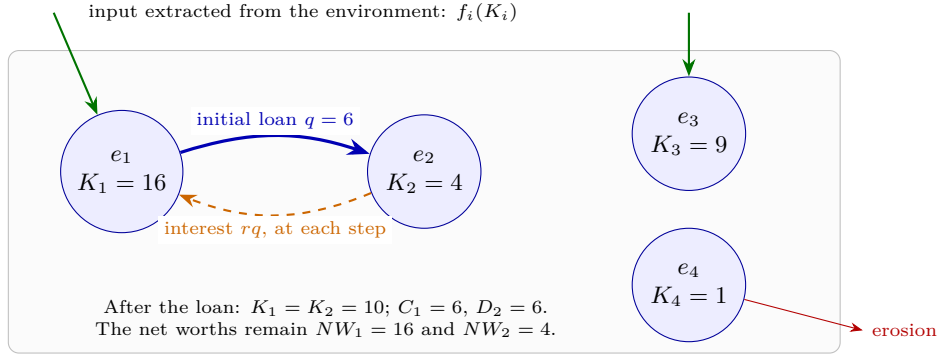


Figure 1: Real stock and accounting commitment are not the same thing. Fictitious example, inspired by the pedagogical diagram of the internship presentation. The initial values are chosen to explain the mechanism, not extracted from a simulation. Entities 3 and 4 show that not all entities have to take part in the same exchange. Inputs and erosion are drawn on only a few entities to lighten the diagram; the rules apply to all. The external reservoir is not simulated.

Object	Dimension and operation	Proposed economic reading
K_i	Stock, in model joules	Mobilizable productive size
$A_i K_i^{\gamma_i}$	Production added during a step, in joules	Productive flow, not a wage
$A_i K_i^{\gamma_i} / \Delta t$	Mean power over a step	Production capacity per unit of time
q	Nominal principal, same unit of account as K	Claim and debt, not amortized
rq	Transfer during a step	Interest service, not resource creation
NW_i	Accounting stock $K_i + C_i - D_i$	Balance-sheet position, distinct from the physical stock

A watt is one joule per second. The engine's outputs are in joules *per step*; calling them watts would require fixing Δt in seconds, which has not been calibrated.²

The interpretive hypothesis is that certain monetary transfers represent transfers of entitlements to mobilize exergy. The real stock and the nominal network are thus coupled, but not identical: a loan moves K and simultaneously records a claim and a debt. Production represents an input to the modeled stock; the external reservoir that enables it is not described. The internal balance sheet is closed in accounting terms, not thermodynamically over the whole economy–environment system. Furthermore, the observable called "total income" is $\text{prod} + \text{int_in}$: it adds production and transfers received, and must not be equated without correction to an aggregate net production.

²In the update $K \leftarrow K + AK^\gamma$, A has dimension $J^{1-\gamma}$ for the production of one step. The coefficient of a corresponding differential equation would have dimension $J^{1-\gamma}/s$. Comparing them without specifying the step would make a physical dimension disappear. The rate r is a fraction due per step; $r/\Delta t$ has the inverse dimension of a time.

2.5 Scales: which parameters are truly independent?

In the homogeneous regime, without noise or credit, the order production then depreciation gives

$$K_{t+1} = (K_t + AK_t^\gamma)(1-\delta), \quad K_{\text{aut}} = \left[\frac{A(1-\delta)}{\delta} \right]^{1/(1-\gamma)}$$

for $A > 0$, $0 < \delta < 1$ and $0 < \gamma < 1$. K_{aut} is the stock toward which an isolated entity, without credit or noise, tends when extraction and depreciation offset each other. This deterministic fixed point is not a mean of the noisy system. Setting $x_i = K_i/K_{\text{aut}}$ transforms production into

$$x_i \leftarrow x_i + \frac{\delta}{1-\delta} x_i^\gamma.$$

At fixed institution, the common scale of A and K° can thus be replaced by the ratio $\kappa^\circ = K^\circ/K_{\text{aut}}$: it measures the size of a new entity relative to what it could reach alone. A low value means it enters far below this scale. For a toy model, avoiding double-counting the same scale freedom is essential.³

The rescaling $K, q, K^\circ \mapsto c(K, q, K^\circ)$ and $A \mapsto c^{1-\gamma} A$ preserves the homogeneous equations and the marginal rates. Absolute numerical thresholds and the engine's rounding limit software exactness; recalibrating the time step is not an exact symmetry of the discrete market: the numerical study is presented in §5.4 and the non-equivalence of the operations is detailed in appendix G.4. A rise in A at fixed K° therefore changes κ° , whereas its compensation $K^\circ \mapsto K^\circ m^{1/(1-\gamma)}$ preserves it. The two are distinct experiments: compensation is a control, not an imposed physical correction.

The tension $T = K_{\text{aut}}/K_{\text{eq}}$, with $K_{\text{eq}} = (Y/(N_{\text{prod}}A))^{1/\gamma}$, compares the individual scale without credit to the observed productive scale. Here Y is the total production of a step and N_{prod} the number of

³An equivalent option is to choose K° as the unit and to retain $a = A(K^\circ)^{\gamma-1}$. The point here is to eliminate *one* unit-of-capital freedom, not to show that the other parameters are redundant: $\gamma, \delta, \sigma, \lambda$ and the institution remain relevant. An invariance of certain statistics with respect to λ is not a trajectorial equivalence of populations of different sizes.

entities present in the productive phase, which can differ from the surviving headcount at the end of the step. K_{eq} is the fictitious stock that would give a representative entity the mean production Y/N_{prod} ; for $\gamma \neq 1$, this is not the mean stock. A large T therefore indicates a productively equivalent stock far below the autarkic scale. This tension is useful in particular at fixed δ ; it is neither a universal state variable, nor another name for the whole credit–mortality–population chain. The coupling of distinct physical and economic spheres joins an existing thermodynamic motivation [6], without reusing its model or identifying money with exergy.

2.6 Credit as cooperation, and the rate as sharing

For two entities with the same concave technology, the marginal return $f'(K) = A\gamma K^{\gamma-1}$ is larger for the less endowed one. A transfer toward it therefore increases *joint* production as long as it does not exceed the equalization of marginal returns. This result does not hold for any amount nor for any heterogeneous pair. It is the sole motive for credit in this model: there is neither anticipation, nor time preference, nor price.

Figure 2 illustrates the role of technology and the gain from lending. At fixed γ , A multiplies extraction per step; on its own it is not a power in watts. The exponent γ measures individual returns to scale. Between zero and one, it makes the function concave and allows a reallocation gain: calling it directly a "propensity to collaborate" would be excessive. At $\gamma = 1$ and identical technology, this gain disappears; its variation with γ requires fixing the units and the states compared.

The market phase carries out $\lfloor \rho n \rfloor$ rounds for an eligible pool of n entities in the configurations studied; ρ is the market intensity (default value 1). Encounters are pairwise. The amount targets the equalization of marginal returns in the main configuration. The free direction of the loan recovers the rich-to-poor transfer in the homogeneous regime, not necessarily in the heterogeneous regime.

There is therefore a *cooperation surplus* to be shared, and the interest rate is the sharing mechanism. For a contract of principal q , let L denote the production loss suffered by the giver by parting with this stock, and Δ the joint production gain of the pair – the surplus. The service paid per step then decomposes as

$$r q = L + p \Delta. \quad (3)$$

The number p is the *sharing*. The imposed *bargain* rule explores $p \in [0, 1]$; the implicit sharing of another rule can exceed these bounds. At $p = 0$, the service exactly compensates the productive loss computed at contract creation; the receiver keeps the surplus. At $p = 1$, the lender also receives all of this surplus. This instantaneous comparison does not guarantee the contract's balance over its lifetime. Equation (3) defines p for any rate rule, which makes it possible to place a given rule on the sharing scale – this is what we exploit in §7. The reference rule, used by default, sets r to the geometric mean of the marginal returns of the two entities.

2.7 Bankruptcy and cascade

An entity goes bankrupt if it is in service default or if $NW_i < 0$. The rule is deliberately brutal: its contracts are *cancelled* – its creditors lose the principal, with no recovery or buffer – and its residual capital is *destroyed*. A default therefore directly annihilates assets held by others and can render them insolvent in turn; the new insolvencies form the next generation, and iteration continues to the fixed point *within the same step*.

This rule is what distinguishes a contagion from mere synchronization. When a borrower dies, each lender loses the corresponding principal, which defines a directed edge from the dead entity to the lender. An *avalanche* is a weakly connected component of this loss graph, restricted to entities that died in the same step.

Two propagation measures follow: the *branching ratio* $b_1 = 1 - (\text{roots/deaths})$, where a root is a bankruptcy that entered directly in the step's initial queue, and the *mean offspring* b_2 , the mean number of causal pairs between successive generations per death. The two coincide if and only if every non-root death has exactly one parent; their gap measures the excess of edges per death: $b_2 - b_1 = \sum_v (d_v - 1)/N$, with d_v the number of previous-generation parents of a non-root and N the total number of deaths. This is neither the proportion of victims with several parents, nor the number of individually sufficient causes.

3 Experimental approaches in several steps

The first step is a *mapping*: varying parameters to distinguish robust regularities from effects specific to a given configuration. The M4B study explores births, stock erosion, noise, endowment and the size of the encounter pool. It includes one-parameter-at-a-time variations, a global plan of 64 points with three seeds, two-parameter cuts and a confirmation with five new seeds. Its inventory counts 594 simulations, not all of whose results are reproduced here. The confirmation center is $\lambda = 30$, $\delta = 0.05$, $\sigma = 0.25$, $K^\circ = 25$, $k = 3$, $A = 1$, $\gamma = 1/2$.

The second step is *scale identification*: determining what amounts to a freedom of unit and what actually changes the exchanges. Time-rescaling trials use three seeds per configuration, with durations brought back to a common unit. They are presented separately, since their protocol is not that of the technological interventions.

The third step is the *paired intervention*. At the main point, $\lambda = 30$, $K^\circ = 25$, $\sigma = \delta = 0.01$, $A = 1$, $\gamma = 1/2$, $\rho = 1$ and $k = 2$. The population numbers about 1132 entities in the observed regime. Twelve states are seeded up to step 2000; each yields a control and variants, with the same initial state and the same generator at the branch point. Individual draws can then diverge with the population sizes. The M4.4 campaign comprises 372 inventoried simulations, not 372 independent repetitions of a single experiment.

The M4B results and those of the main configuration are not interchangeable: beyond noise and depreciation,

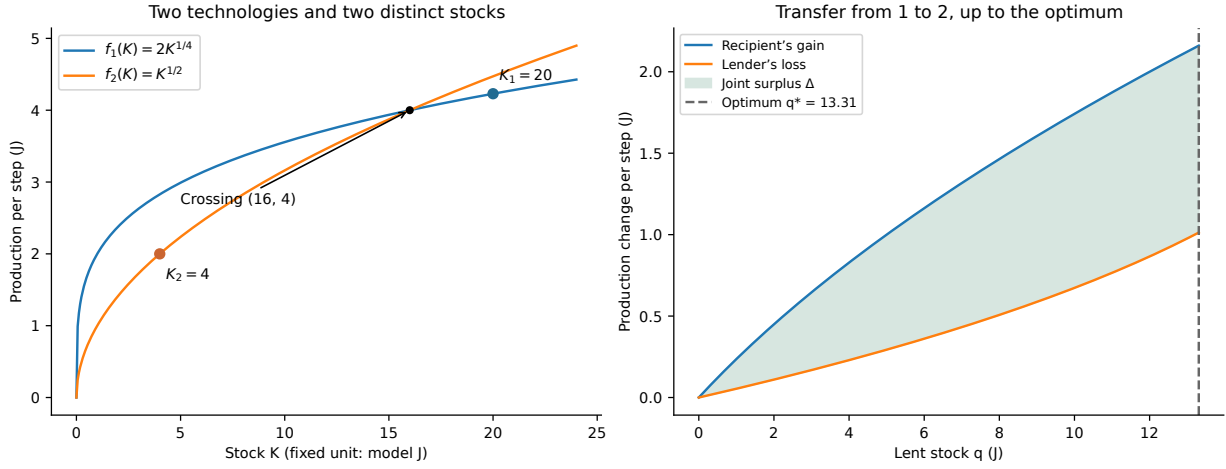


Figure 2: Technology and cooperation gain. On the left, the technologies $f_1(K) = 2K^{1/4}$ and $f_2(K) = K^{1/2}$ cross at $K = 16$, for a production of 4 per step; the initial stocks are distinct, $(K_1, K_2) = (20, 4)$. The coefficients are expressed in a single fixed stock unit. On the right, a loan of 1 toward 2 increases joint production up to the equalization of marginal returns. The area between gain and loss measures the joint surplus; the crossing of the productions on the left is not this equalization. These curves are analytical, at fixed states, without a statistical interval.

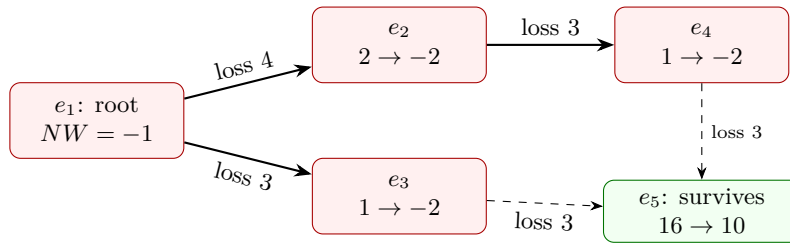


Figure 3: A directed propagation tree, not the direction of the loan. Example that is accounting-feasible with five entities. Before resolution, their stocks are worth $(6, 1, 1, 1, 10)$; the loans are $e_2 \rightarrow e_1 : 4$, $e_3 \rightarrow e_1 : 3$, $e_4 \rightarrow e_2 : 3$, $e_5 \rightarrow e_3 : 3$ and $e_5 \rightarrow e_4 : 3$. The arrows in the diagram go, conversely, from the defaulting borrower to the affected creditor. The component of the four deaths has one root and two generations of propagation: $S = 4$, $b_1 = b_2 = 3/4$. The losses suffered by e_5 , a survivor, do not make it part of the avalanche of deaths. The general model allows several parents, unlike this tree-shaped example.

M4B uses a pool of k entities and a geometric target, while the main configuration uses pairwise encounters and a marginal-equalization target. Each figure states its family, its window and its number of seeds. Engine names serve to trace sources, not to replace these scientific definitions.

The variation of k belongs to the historical M4B exploration; the later main experiments fix encounters at $k = 2$. This choice removes k from the levers of this configuration, without retroactively modifying the parameters of the earlier experiments. Software extensions have undergone compatibility checks in specified regimes; this does not make different market rules equivalent. Appendix G.3 distinguishes these checks from changes to the model.

A seed, not an entity or a step, constitutes the unit of repetition. Unless stated otherwise, levels are averaged over $3000 < t \leq 4000$, then ratios are computed seed by seed. A 95% CI bar is a Student interval between these repetitions. A drift diagnostic compares the first and last quarters of the window; its formula, its results and its limited scope are given in Appendix C. The absence of a detected mean drift does not demonstrate strict stationarity.

4 What emerges without being imposed

4.1 A renewed population and a credit network

The rules fix the mean flow of births, not the population size, the lifetime or the lender/borrower roles. In the configurations retained, the series settle into a fluctuating regime with no observed unbounded growth. Entries balance exits on average, but a stable age distribution does not imply that any particular entity is at equilibrium. New entities grow, incur commitments and disappear at varied ages.

Little’s law relates, under a stable regime and with controlled temporal boundaries, the mean population to the entry flow multiplied by the mean lifetime. It is a demographic identity useful for organizing measurements, not an explanation of the lifetime. The lifetime depends on noise, the network and the bankruptcy rules. The deterministic equilibrium of an isolated entity does not by itself predict the collective state.

On a single realization, this renewal becomes directly readable. Figure 4 shows a 300-step window taken from the stationary regime of the main control, seed 0 ($3400 < t \leq 3700$): population and total productive stock fluctuate in a visibly correlated way, and new loans form at every step without any net trend settling in at this scale. This reading concerns a single seed; it makes the mechanism tangible, it does not replace the between-seed statistics given afterward.

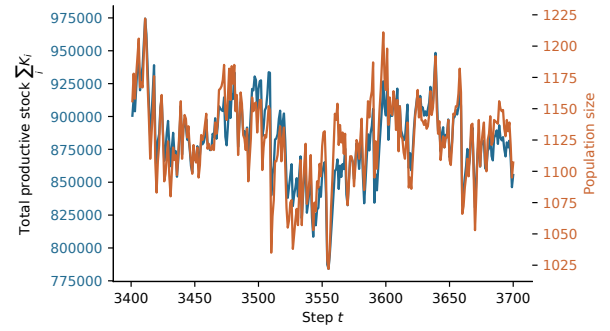


Figure 4: A window of the raw trajectory. Main control, seed 0, $3400 < t \leq 3700$, no smoothing. Population size (right axis) and total productive stock $\sum_i K_i$ (left axis), step by step. Window chosen to stay within the stationary regime (Appendix C), with no particular property sought a priori.

4.2 Directed cascades and a wide range of sizes

The loss graph makes it possible to distinguish propagation from temporal coincidence. The example in Figure 3 gives the definition; the distributions in Figure 5 show the simulated realizations. We fit, on each seed, a discrete probability

$$P(S = s) = Z^{-1} s^{-\alpha} \exp(-s/S_c), \quad s \geq 1, \quad (4)$$

where Z normalizes the sum of probabilities and S_c describes a soft cutoff, not an absolute maximum size. At the main control, the twelve fits give $\alpha = 1.623 \pm 0.009$ and $S_c = 206 \pm 11$; under $A \times 1.5$, $\alpha = 1.667 \pm 0.009$ and $S_c = 185 \pm 7$. These uncertainties describe the between-seed variability of this estimator. They validate neither the chosen family nor the identification of the cutoff with the system size.

The cutoff model is fitted by likelihood on integer sizes, not by selecting the straightest portion of the histogram. The mean R^2 values, about 0.938 and 0.928, do not constitute a test of Pareto behavior. Despite these high values, the curves underestimate part of the intermediate sizes and overestimate the extremes: the visible deviations exceed several of the between-seed bars. The simple form of Equation (4) therefore does not account for the whole distribution at the precision of the data. Choosing the cutoff to simultaneously optimize a bootstrap variance and an R^2 would also select favorable fluctuations; we retain here one stated model and support, with the caveats of Appendix D.

At the main point, $b_1 = 0.7863 \pm 0.0013$ and $b_2 = 0.8483$: some deaths have several parents. The fact that $b_1 < 1$ is not, by itself, a test against criticality: its definition as the fraction of propagated deaths bounds it by construction. Conversely, an extended tail does not demonstrate SOC. Finite-size study and the definition of events across several steps remain necessary.

This pointwise average masks a size dependence. Figure 6 breaks down, over the 4086 avalanches of the main control within the stationary window (seed 0), the propagation depth and the share of non-root members by size class. Size-1 avalanches are, by construction, iso-

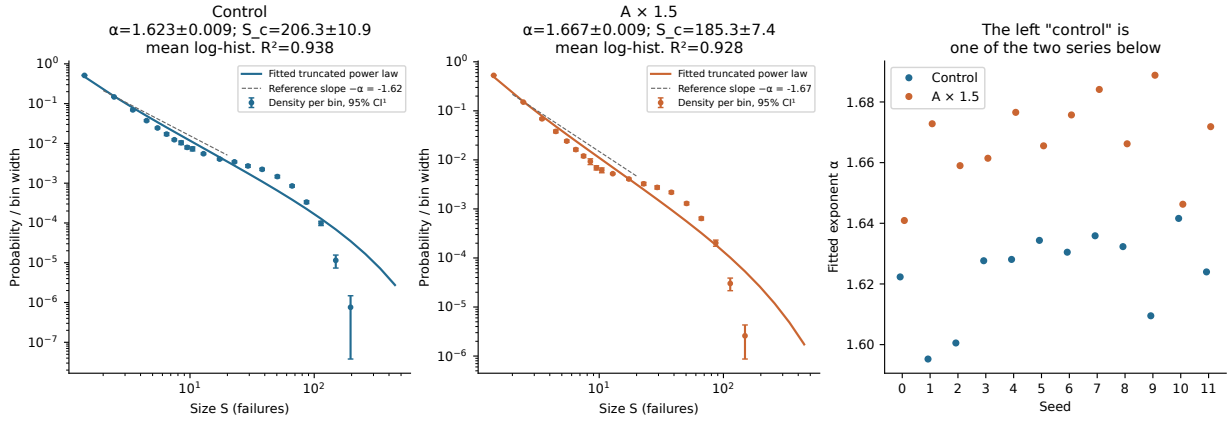


Figure 5: Histograms, fits and repetitions. M4.4, twelve seeds per condition, $3000 < t \leq 4000$. Points give the per-class probability divided by class width; bars are between-seed 95% CIs. Curves are the means of the discrete cutoff-exponential fits, ribbons a bootstrap of 2000 draws of whole runs. Dashed lines indicate the reference slope $-\alpha$, not the complete truncated law. R^2 values are computed on the logarithms of non-empty classes, then averaged across seeds. The left panel represents the control whose individual exponents are repeated on the right; the central panel represents the other series. ¹Bars too long for the log scale are truncated at 95 % of the level; their full values remain in the data.

lated roots with no depth; beyond that, both depth and non-root share grow with size, with no sharp threshold separating two regimes.

4.3 Short contractions, distinct from avalanches

An avalanche is a network event within a single step. A contraction is a sequence of steps over which an aggregate quantity decreases. They are therefore neither the same observation nor two definitions of the same size. In the central M4B regime, the mean durations of stock and production contractions are close to two steps. Figure 7 formalizes their decay by a discrete exponential, $P(D \geq d) = \exp[-\beta(d-1)]$, $d = 1, 2, \dots$

We find $\beta = 0.731 \pm 0.029$ for stock and 0.696 ± 0.035 for production, with mean log-survival R^2 close to 0.98. Wright [8] shows precisely why power law and exponential must be compared over the whole set of durations, without discarding the shortest episodes. His study covers annual GDP contractions of 17 economies between 1870 and 1994; our uncalibrated steps are not years. The resemblance in shape motivates a future comparison; these short contractions do not by themselves support a power law of durations nor the SOC hypothesis.

4.4 Balance and interest distributions: what plausibility?

Net worth (denoted NW) is the model's quantity closest to a net wealth: real assets plus claims minus debts. Interest received constitutes a financial income flow. Their distributions emerge from entities subject to the same rules, not from wealth classes imposed at birth. In the main regime, the time-averaged Gini indices are close to 0.613 for net worth and 0.513 for interest received. Figure 8 represents distinct snapshots of these time averages.

To gauge orders of magnitude, the net-worth Gini of US households is 0.852 in the 2019 SCF [9]: the main

model is thus noticeably less unequal on this measure. The CBO reports a Gini of 0.516 for income before means-tested transfers and taxes in 2019 [10]. Its numerical closeness to 0.513 does not validate the model's distribution: interest received includes neither wages nor all capital income, and our entities are not households weighted by size. A direct confrontation requires the same type of actor, the same period and the same accounting definition. We therefore do not compare the Gini of physical stock to that of income to conclude on economic plausibility.

The tails of net worth and interest are compatible with power-law fits over part of the domain; their statistical identification remains open [5, 7]. A log-normal and a cutoff power law each have two shape parameters at a fixed threshold. If the cutoff depends on system size, that dependence must be estimated across several sizes; it does not automatically remove a parameter from a fit performed at a finite size. The existing tests and their limitations are detailed in Appendix D.

5 Steering: which levers change which properties?

5.1 A hierarchy of sensitivities, not a single control knob

The M4B mapping separates effects that would be confounded in a comparison of only two trajectories. Noise σ affects survival and exposure; endowment K° changes the entry position of young entities. Depreciation shifts the productive scale and also enters balance sheets; the birth rate λ allows exploring populations of different sizes. These are tendencies within the domain studied, not exclusive roles nor demonstrated independences.

The finite-size variation is particularly important for the critical hypothesis. In the M4B confirmation, as λ goes from 10 to 30 then 100, the mean observed maximum avalanche size goes from 29 to 54 then 110.

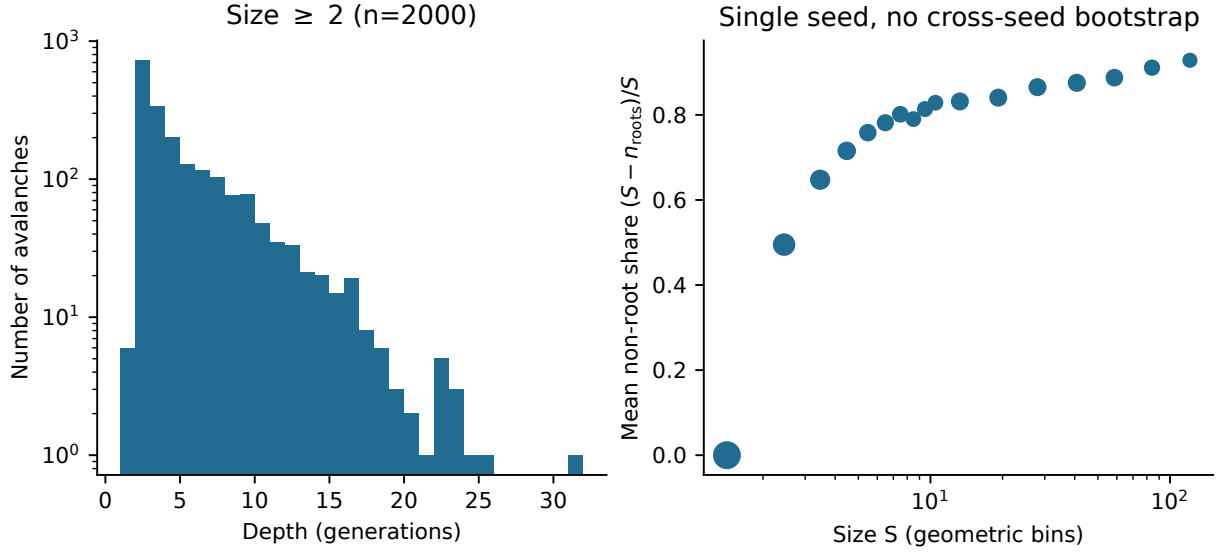


Figure 6: Cascade shape by size. Main control, seed 0, $3000 < t \leq 4000$, 4086 avalanches. Left: histogram of propagation depth (number of generations), log scale, avalanches of size ≥ 2 only. Right: mean share of non-root members $(S - n_{\text{roots}})/S$ by size class (geometric bounds), number of avalanches per class shown; no between-seed interval for a single realization. The pointwise b_1 value cited in the text is the avalanche-weighted mean over the whole window, not the mean of the points in this panel. A single seed; descriptive reading, not a new fit.

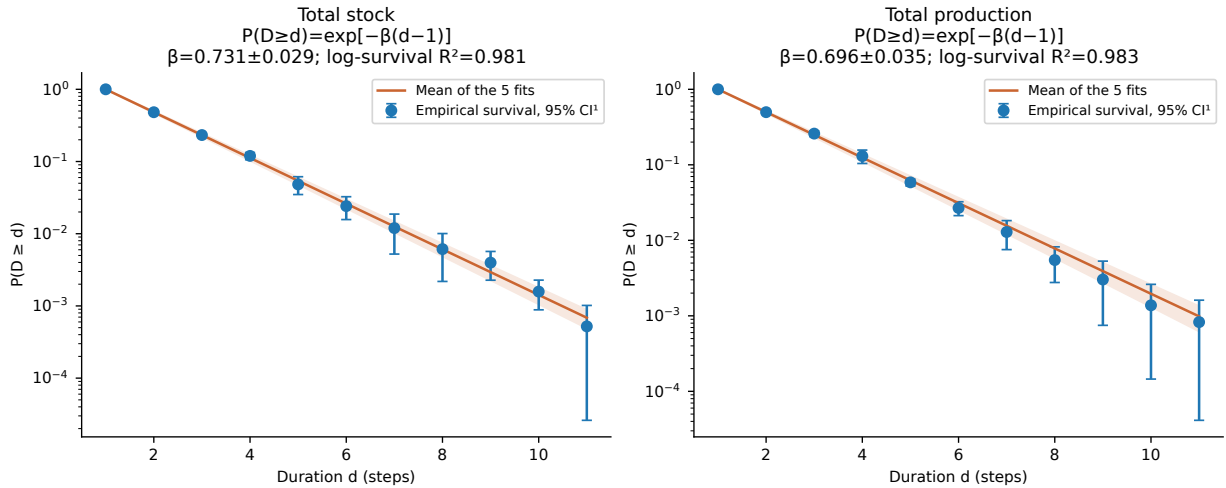


Figure 7: Contraction durations: an exponential, not an imposed power law. Central M4B, five seeds 11–15, $1000 < t \leq 4000$, unsmoothed; stock and production are treated separately. β is estimated per seed from the mean duration. Points are the empirical survival values and their 95% CIs; curves average the five fits. Descriptive R^2 values bear on the log-survival at the observed durations, whose points are dependent. Logarithmic bars are truncated according to the convention of Figure 5. Episodes touching a window boundary are counted per the existing protocol; this is not a survival analysis correcting for censoring.

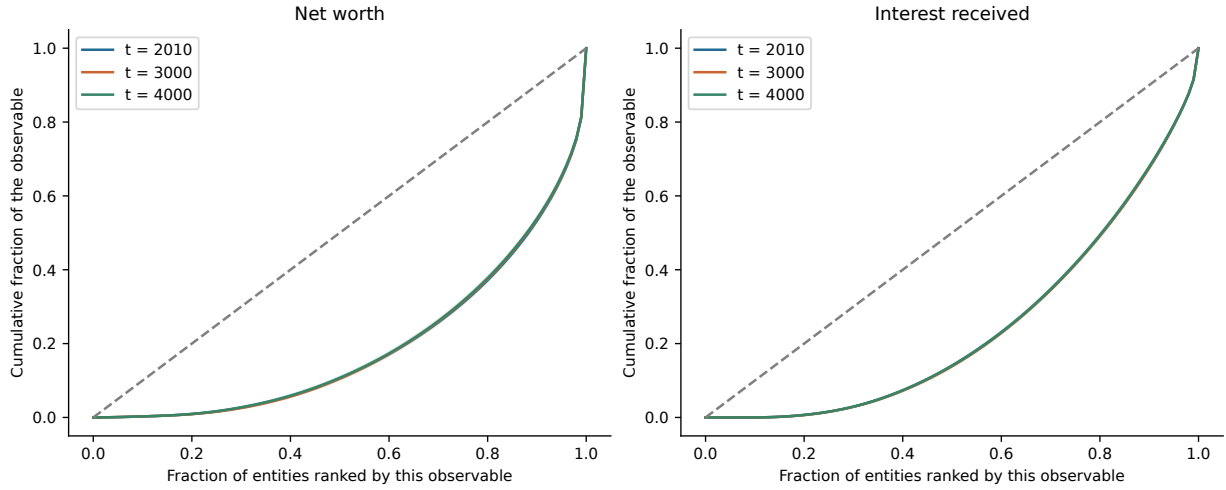


Figure 8: The collective distribution persists despite individual turnover. M4.4 control, twelve seeds: Lorenz curves at $t = 2010$, 3000 and 4000, and between-seed 95% CIs. Each observable has its own ranking.

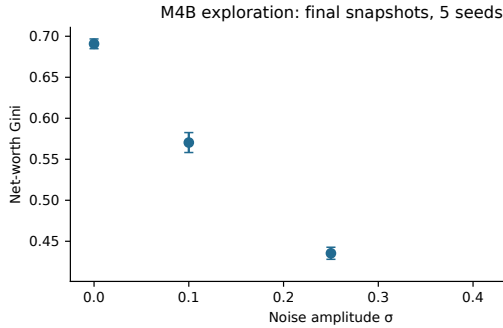


Figure 9: Returning to the levers studied during model construction. M4B confirmation, five independent seeds per cell, $T = 4000$. Net-worth Gini computed on the final snapshot, not the Gini of capital nor of the pool. The other parameters are those of the historical M4B center, notably $k = 3$; only σ varies. The old sweep of k is not shown here, as the later main experiments fix encounters at two entities. Bars are Student 95% CIs recomputed from the per-seed values.

The fitted cutoffs do not all behave as regularly: at $\lambda = 100$, the estimated value is far beyond the sample, hence poorly identified. These observations support a finite-size study; they do not justify imposing a universal cutoff law or mixing the M4B and M4.4 exponents.

5.2 Encounter intensity as an experimental lever

In the main configuration, ρ sets the number of encounters per step. Varying it from 0.5 to 3 allows measuring its response at five levels, with twelve seeds per level. The fitted exponents of the interest and net-worth tails, as well as branching, increase; the net-worth Gini varies more weakly.

The measured log-log slopes are 0.0465 ± 0.0074 for the interest exponent, 0.0272 ± 0.0074 for the net-worth exponent and 0.1345 ± 0.0020 for b_1 . That of the net-worth Gini is -0.0084 ± 0.0023 : a weak but detectable effect on average, which should not be confused with the

strong response of the pool's Gini. A significant mean slope does not imply that every increment is monotone within every seed. The result is a local steering lever, not a universal law nor proof of a distribution family.

5.3 Changing a scale is not always changing the system

Non-dimensionalization (§2.5) removes a freedom of unit: multiplying all stocks and principals by c and A by $c^{1-\gamma}$ preserves the homogeneous equations. The quantity $\kappa^\circ = K^\circ / K_{\text{aut}}$ instead expresses a relative position that changes the dynamics. At $\gamma = 1/2$, multiplying A by 1.5 multiplies K_{aut} by 2.25; keeping K° fixed therefore divides κ° by 2.25.

This distinction organizes the experiments: a technical change at fixed endowment tests a relative displacement of new entrants, while a compensated endowment preserves that relative position. One must further distinguish compensating only the births after an intervention from rescaling *the whole* initial state, including the nominal network. Only the latter directly falls under the exact symmetry of the equations.

5.4 The granularity of the time step

Depreciation represents a temporal erosion. If a step lasts Δt and the continuous erosion rate is d , then $\delta = 1 - e^{-d\Delta t}$. Grouping s steps therefore requires

$$\delta' = 1 - (1 - \delta)^s, \quad \sigma' = \sqrt{s} \sigma, \quad \lambda' = s\lambda.$$

The last two relations compose, respectively, independent log-normal shocks and Poisson births *in law*, for isolated phases. They do not guarantee that the coupled system follows the same law. Per-step flows must also be converted: $A' \simeq sA$ and $\rho' \simeq s\rho$; the nominal service of existing contracts must be converted if one transforms an already indebted state.

The simulations of Figure 11 leave a residual even after converting all these coefficients. Two steps successively update production, commitments, population and encounters; a single more intense step does not reproduce

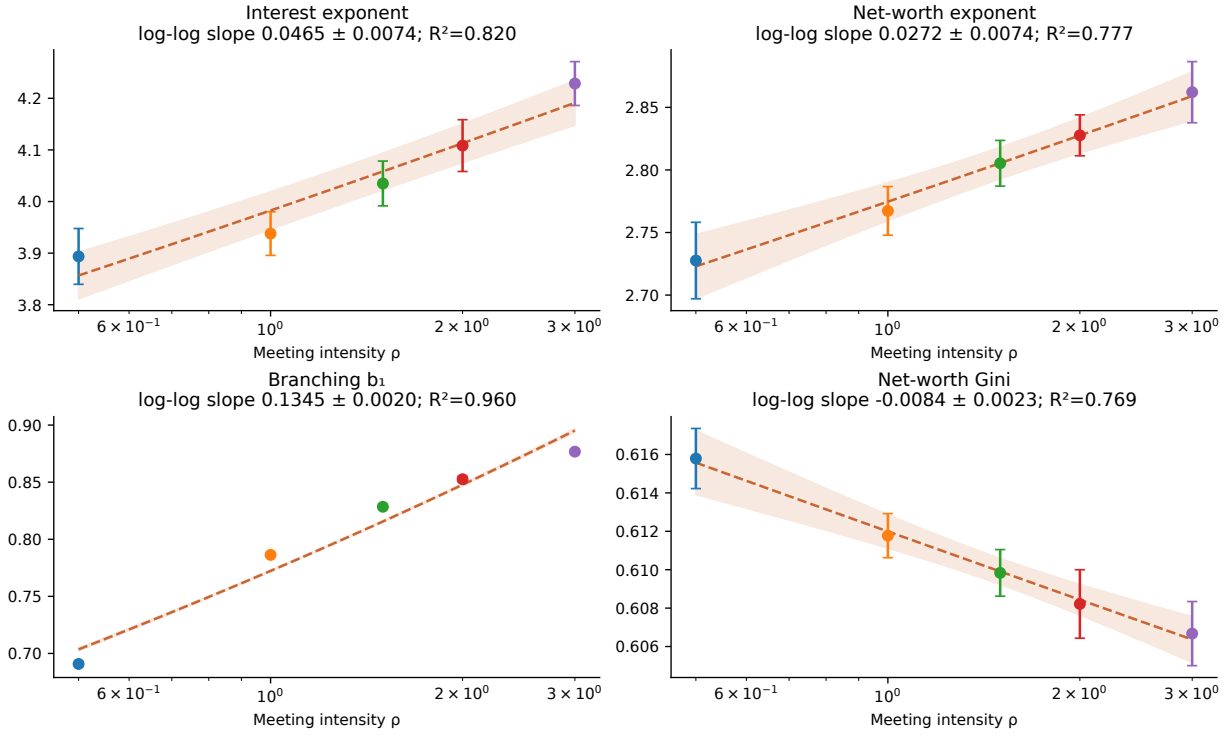


Figure 10: Measuring steering on defined observables. M4.4, five values of ρ , twelve seeds, $3000 < t \leq 4000$. Exponents and branching are those of the per-seed analyses; the net-worth Gini is recomputed on the panels of the same window. Points: means and 95% CIs. Curves: per-seed log-log fits; slopes and R^2 are then averaged, with 95% CIs of the slopes.

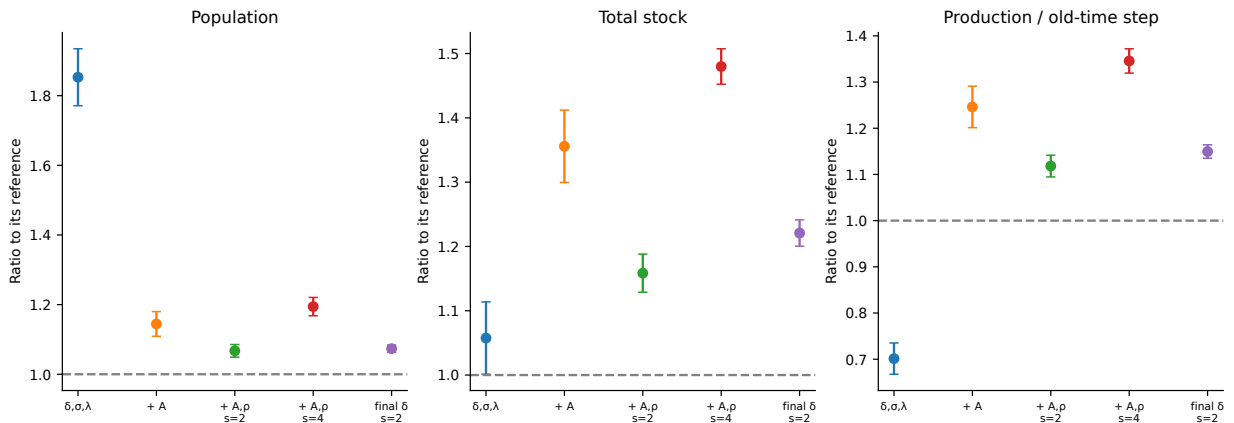


Figure 11: Changing the step does not stop at three substitutions. Live-v1 experiments from an empty state, three seeds per configuration. Common window $1200 < t_{\text{ancien}} \leq 2000$. Flows of the grouped steps are divided by s , not the stocks. The first three trials group two steps; the fourth groups four. The last compares two configurations at $\delta = 0.002$ before conversion, and therefore has its own reference. Points: per-seed averaged ratios; bars: Student 95% CIs, wide with three repetitions.

this alternation. Reducing δ alone does not remove the residual in the trial performed. These controls rule out a simple unit error as a complete explanation, without isolating each compositional effect. A proper convergence study would require jointly reducing all relevant increments and comparing at a fixed physical horizon. It remains to be carried out.

6 Collective response to a technological improvement

6.1 What changes and what the elasticity measures

The main experiment multiplies A by $m = 1.5$ for all living entities and for future births, starting at step 2001. At unchanged stock and fixed exponent γ , each entity would produce exactly 50 % more: this is the proportional reference. But stocks, the network and the population size then evolve. We measure

$$\varepsilon_s = \frac{\ln(\bar{Y}_{s,T}/\bar{Y}_{s,C})}{\ln(1.5)}, \quad \hat{\varepsilon} = \frac{1}{12} \sum_{s=1}^{12} \varepsilon_s, \quad (5)$$

where Y is total production, T the treated arm, C the control and the overbar a mean over $3000 < t \leq 4000$. The denominator is the logarithm of the imposed technical factor, not a fitted constant. A proportional response gives $\varepsilon = 1$; $0 < \varepsilon < 1$ means a sub-proportional increase; $\varepsilon < 0$ would mean a decrease despite the improvement. This is a *finite* elasticity, not an infinitesimal derivative.

6.2 A positive but non-proportional response

At fixed birth endowment, $\hat{\varepsilon} = 0.7473 \pm 0.0106$. **The collective response is not proportional to the individual improvement.** The representative factor $1.5^{0.7473} \simeq 1.354$ corresponds to about 35 % additional production. The arithmetic mean of the ratios is not exactly this transform of the mean of the logarithms; the per-seed values are kept.

The end-of-step population contracts by about a quarter, while production per entity increases. To keep an exact decomposition after time averaging, one can define the mean population yield by $\bar{y} = \bar{Y}/\bar{N}$. Then $\ln(\bar{Y}_T/\bar{Y}_C) = \ln(\bar{N}_T/\bar{N}_C) + \ln(\bar{y}_T/\bar{y}_C)$ by definition. $\bar{y}$ is not in general the time average of $Y(t)/N(t)$; one must also distinguish the step's producers from the end-of-step survivors. A decomposition identity does not constitute a demonstration of the demographic mechanism.

If instead the endowment tracks the productive scale, the measure is 2.0029 ± 0.0148 . This value is compatible with $1/(1 - \gamma) = 2$, the prediction of full homogeneous rescaling, but numerical compatibility does not make the two protocols identical. Precisely, at $\delta = 0.01$ and $\gamma = 1/2$ fixed, K_{aut} goes from 9801 to 22052.25 when A goes from 1 to 1.5. At unchanged endowment $K^\circ = 25$, the ratio κ° goes from about 0.00255 to 0.00113: new entities enter smaller relative to this scale. Compensation

consists in raising their endowment to 56.25, in order to preserve κ° , not to cancel the production increase or compensate a loss. It increases the real contribution per birth; the stocks of already present entities and their principals are not rescaled. The two arms thus compare different entry conventions, not the same energy intervention at unchanged contribution. **In both experiments production increases; it is the deviation from proportionality that changes sign.** A negative response to an increase in A has not been established; its exploratory search is not presented here as a result.

6.3 An explanatory chain to test link by link

At fixed endowment, the improvement reduces κ° : new entities enter smaller relative to the productive scale. The data simultaneously show a variation in the dispersion encountered on the market, in credit volume and in mortality. They motivate the chain below, whose arrows do not all carry the same status.

The elasticities of pool dispersion and of credit turnover are 0.5263 ± 0.0069 and 0.5265 ± 0.0067 respectively. The *pool* here denotes the entities drawn for an encounter, not the whole population; *turnover* is the new principal lent per step relative to the total stock. Their numerical closeness is a consistency check, not proof that the first quantity alone causes the second. Mortality per entity responds with elasticity 0.7021 ± 0.0072 ; under a stable regime, it is approximately $\lambda/\bar{N}$.

Branching provides another measure: under $A \times 1.5$ at fixed endowment, the treated-minus-control contrast of b_1 is -0.0322 ; compensating K° leaves a contrast of -0.0007 . Maintaining κ° thus also brings this statistic closer to the control. The decrease in the fraction of propagated deaths does not imply a decrease in individual mortality risk: the two denominators differ.

6.4 Distributions change with cohorts

The aggregate does not say where the distributions move. For each snapshot, we compute the quantiles, then their time average per seed and finally the treated/control ratio. The figure represents them; the tabulated values are in Appendix F.

Production shifts by approximately a factor of 1.8; the extreme quantiles of net worth and gross income increase less than their medians. The first decile of interest received decreases sharply. The compositional control confirms the importance of births: among the 10 % of entities receiving the least interest, more than 99.9 % are at most ten steps old in the control, against about 24 % of the whole population. Their mean age is about 1.8 steps, against 111 steps for the whole population. These ages are measured among the survivors of the main control: $A = 1$, $\gamma = 1/2$, $\lambda = 30$, $K^\circ = 25$, $\delta = \sigma = 0.01$, $\rho = 1$, $k = 2$, free direction and marginal rate. Over the snapshots from 3010 to 4000 spaced ten steps apart, the mean age is 110.97 ± 1.88 steps, and the average of the instantaneous medians is 40.05 ± 0.16

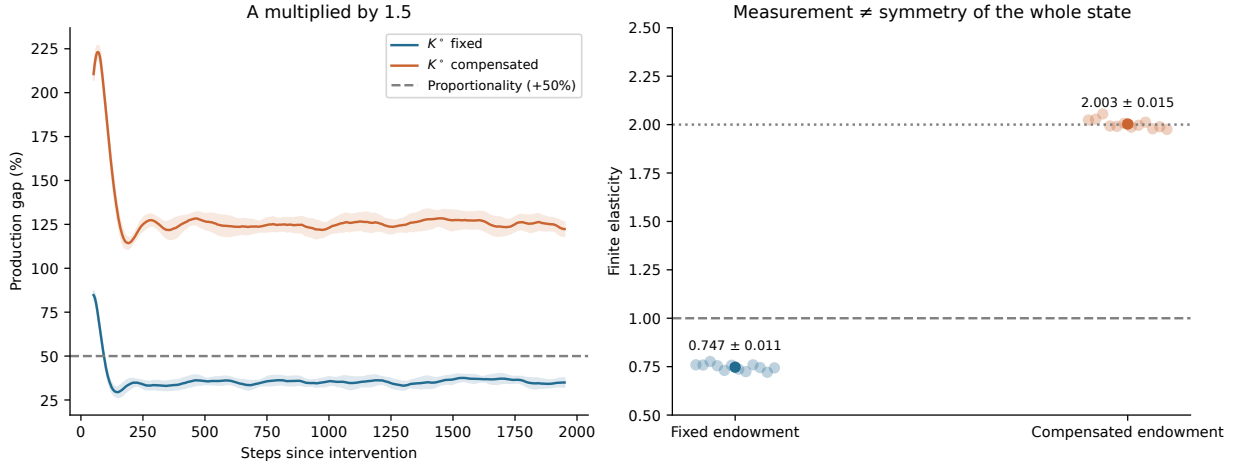


Figure 12: The same improvement, two birth conventions. M4.4, twelve paired seeds. Left, relative treated/control gaps, smoothed over 101 steps per seed before computing the 95% CIs. Right, finite elasticities on $3000 < t \leq 4000$; pale points: seeds; bars: 95% CIs. Compensation applies to births, $K^\circ \rightarrow 2.25K^\circ$, not to already-present stocks and contracts. The line at 2 is a prediction of full-state symmetry, not a constraint imposed on this fit.

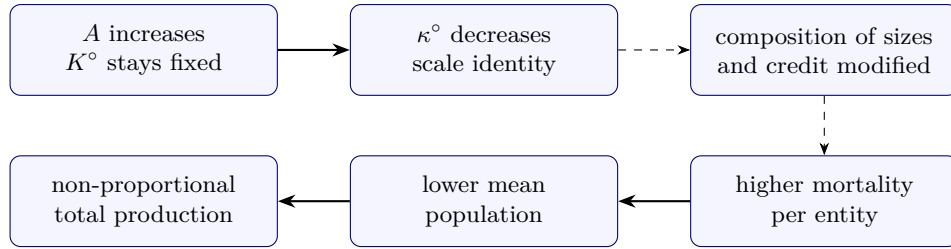


Figure 13: Definitions, measured covariations and proposed mechanisms. The first link is algebraic. The dashed links motivate an interpretation from the paired responses, without isolating a causal mediation. Under a stable regime, mortality per entity and population size are linked by the balance of entries and exits; the last step combines population size with production per entity, which increases. The solid lines do not therefore all denote an experimentally identified causality.

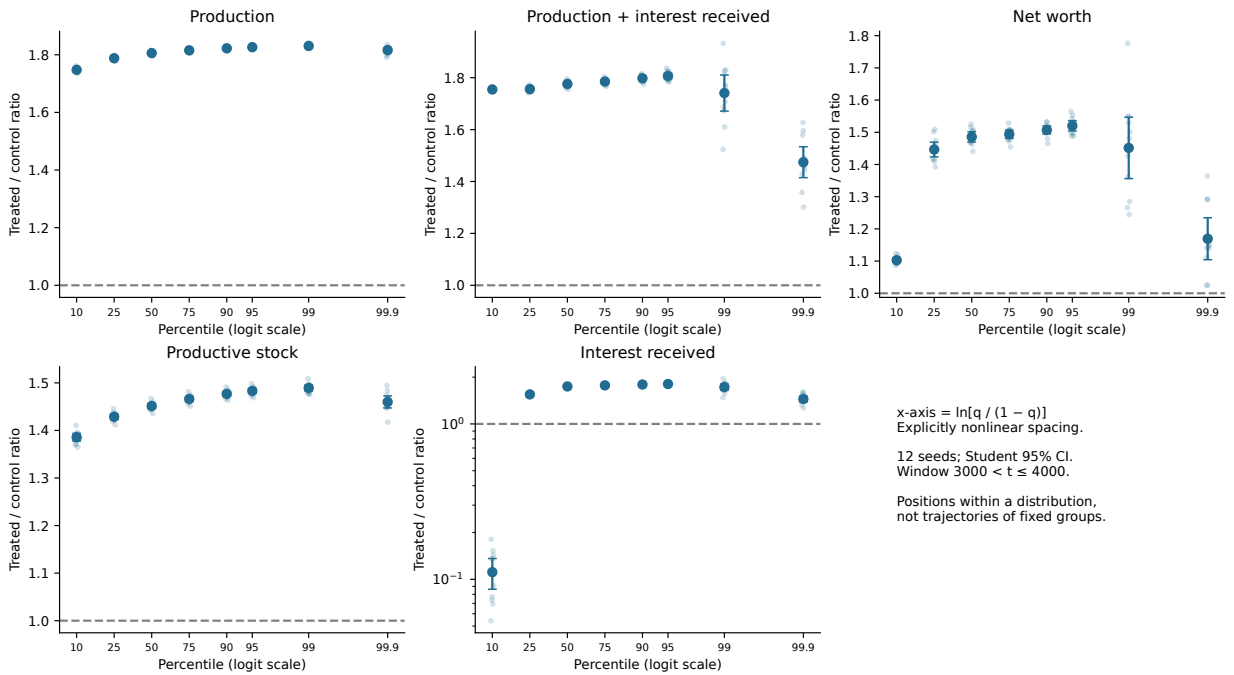


Figure 14: Reading positions, not fixed groups. M4.4, $A \times 1.5$ at fixed endowment, twelve seeds, $3000 < t \leq 4000$. The abscissa is placed at $\ln[q/(1-q)]$, then graded in percentiles: the nonlinear scale is explicit and keeps extreme quantiles legible. Pale points: per-seed ratios; solid points and bars: mean and 95% CI. An ordinate of 1.8 means that the considered quantile is 1.8 times that of the control; it does not mean that the same entities gained 80%.

steps. Each statistic is first averaged in time per seed, then across the twelve seeds; the 95% CIs bear on this latter average. This is neither the median of a sample pooling all instants nor the median lifetime. After $A \mapsto 1.5A$ on the living and future entrants from step 2001, with K° unchanged, these two ages become respectively 94.94 ± 0.86 and 32.03 ± 0.24 steps. A broader exploration of the amplitudes of variation of A and of values of γ remains necessary for this distributional comparison. The young-decile diagnostic is specific to the ranking by interest. In the lowest *capital* decile, the share aged ten steps or less is close to 49 %, not 100 %.

7 The credit institution: a complementary lever

The loan's service allocates the cooperation surplus defined in the model. Imposing a different share p for the lender changes the collective state, without changing the technology or introducing new preferences. This sweep complements the physical steering; it is not the work's leading result.

The population goes from about 1904 to 1138 entities as p goes from zero to one. The reference rule has a mean per-contract share close to 0.5303, but a surplus-weighted share of 0.9412. The difference is not a contradiction: small and large surpluses do not carry the same weight in these averages. The covariance identity and its proof are given in Appendix E.2. It does not establish the causal mechanism of the population change.

Debt relative to production alone does not measure the load ratio on gross income and does not allow ruling out a convexity of risk in this load. Appendix E.1 defines and separately measures the ratio of interest paid to production *plus interest received*. The weighting identity does not exempt one from this risk study.

8 Scope, limitations and next experiments

The model answers a question of existence: can local rules with no rebound behavior produce a non-proportional aggregate response? In the configurations studied, yes. It does not yet settle the share of this mechanism in a real economy. Production added to the stock does not distinguish primary energy, useful service and final consumption; the energy rebound in the applied sense [3] requires these distinctions. The interpretation in terms of exergy entitlements remains a physical hypothesis, not an empirical calibration.

Distributional plausibility is partial: tails emerge, but their families are not established, the Gini levels do not simultaneously reproduce the empirical references, and the "entity" unit is not yet an identified economic actor. Bankruptcy without recovery, perpetual principals and the absence of prices are strong hypotheses. The numerical robustness of the accounts removes none of these limitations.

The SOC hypothesis requires more than a straight-line histogram: it requires an extended finite-size study, a controlled definition of the response to a perturbation, and a treatment of memory across steps. Roots of the intra-step graph may have accumulated fragility in previous steps. Conversely, merging events close in time can artificially create a large component. Temporal granularity and the measurement convention are therefore part of the scientific problem.

The priority extensions are to compare conventions at controlled κ° , to jointly study the heterogeneity of (K°, A, γ) , and to test a finer-step limit. If γ_i varies, the dimension of A_i varies: one must fix a common K_{ref} and a joint law of $(K_i^\circ/K_{\text{ref}}, A_i K_{\text{ref}}^{\gamma_i-1}, \gamma_i)$, rather than drawing values of A_i arbitrarily. An exploratory search for a negative response to an increase in A should state its ranges and windows in advance, separate the transient from the established response, and then confirm any identified configuration on new seeds. No such result is claimed here.

Finally, extending sharing to $p < 0$ would have the lender accept a service initially below its reference productive loss. A later benefit is possible but not guaranteed; it would depend on trajectories and bankruptcies. The engine currently imposes $0 \leq p \leq 1$ for the fixed share: this extension remains an experiment to design, not a regime already simulated.

9 Conclusion

The internship's work builds a device for studying the rebound effect from a physical angle: productive stocks, extraction flows and the network of commitments evolve together. It brings out collective regularities that are not imposed, then searches for the parameters that shift them. Cascade sizes, balance-sheet distributions and contraction durations do not call for a single statistical family; distinguishing them is essential to examining the critical hypothesis.

The response to technology constitutes a result of this program: a uniform individual improvement can produce a sub-proportional collective increase, because the population and its relative positions reorganize. The non-dimensional endowment κ° makes it possible to separate two conventions that here lead to different responses. Time rescaling shows, for its part, that a change of granularity is not a simple symmetry. These results support the interest of an organizational explanation of the rebound, while leaving open the model's criticality and its quantitative confrontation with the real economy.

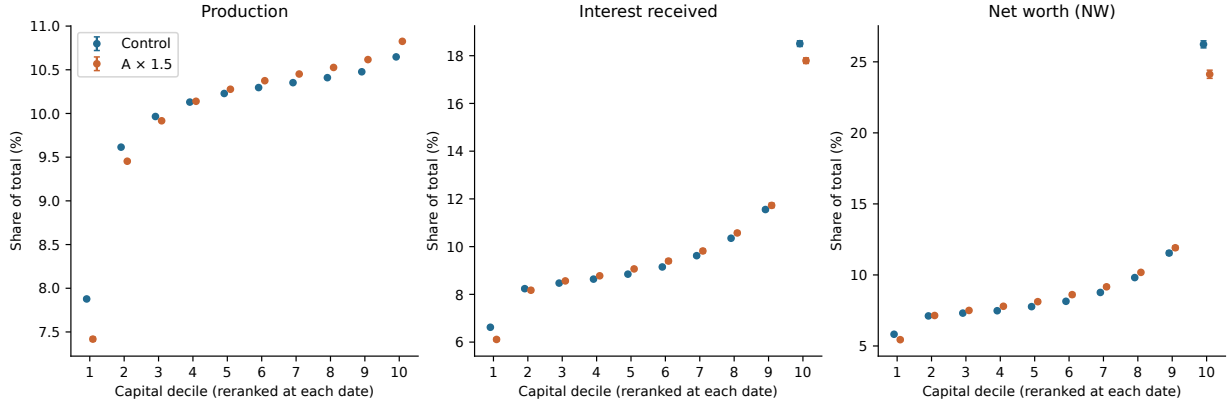


Figure 15: Production, interest and net worth by capital decile. M4.4, same arms and window. Entities are reranked by capital at each snapshot. Reading: the decile-1 point gives the share of production, interest or net worth held by the 10 % least capital-endowed entities at that instant, not by the same individuals over 1000 steps. The strong presence of young entities at the bottom of the ranking contributes to these differences; it is not sufficient to causally explain the whole profile. Net worth is $NW = K + C - D$; it replaces stock in the third panel, without changing the ranking criterion. Groups have equal sizes to within one entity; ties are broken by stable order. Bars: 95% CIs across twelve per-seed means, each computed over the 100 snapshots of the window.

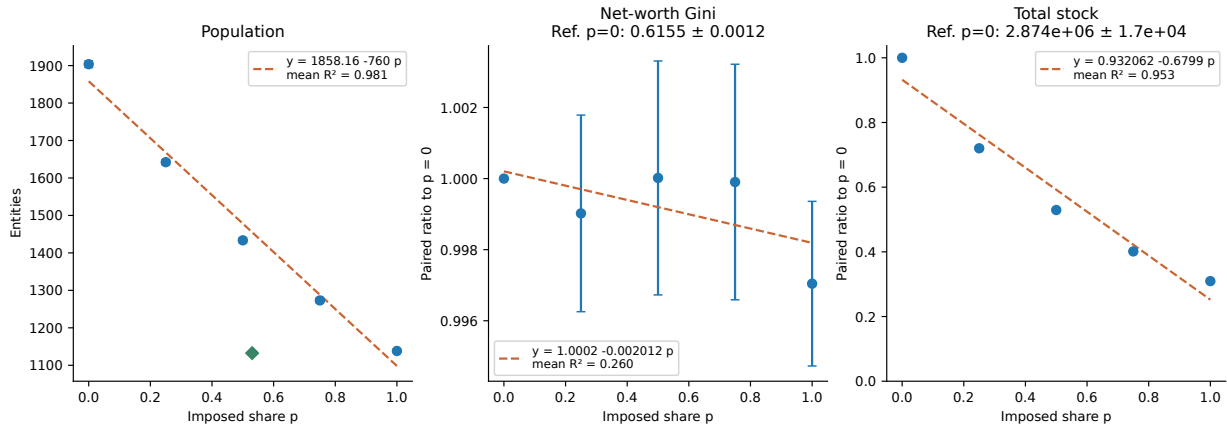


Figure 16: Sharing acts on levels, less on the balance-sheet Gini. M4.4, five imposed shares, twelve seeds, $3000 < t \leq 4000$. Descriptive per-seed linear regressions; mean equations and R^2 shown. Ratios are computed per seed at the $p = 0$ arm, whose raw reference is shown above the panel. The green diamond locates the historical, so-called archaic, marginal rule at its observed mean share. It serves only as an illustrative reference, with uncertainty on both coordinates; it does not enter the regression of the constant shares. A line between simulated levels is not a validated prediction at other values of p .

References

- [1] M. Hendrick, A. Rinaldo and G. Manoli, “A stochastic theory of urban metabolism”, *PNAS* **122**(33), e2501224122 (2025).
- [2] P. Bak, K. Chen, J. A. Scheinkman and M. Woodford, “Self-Organized Criticality and Fluctuations in Economics”, Santa Fe Institute Working Paper 1992-04-018 (1992).
- [3] S. Sorrell and J. Dimitropoulos, “The rebound effect: Microeconomic definitions, limitations and extensions”, *Ecological Economics* **65**(3), 636–649 (2008).
- [4] V. M. Yakovenko and J. B. Rosser, “Colloquium: Statistical mechanics of money, wealth, and income”, *Rev. Mod. Phys.* **81**, 1703–1725 (2009).
- [5] A. Clauset, C. R. Shalizi and M. E. J. Newman, “Power-Law Distributions in Empirical Data”, *SIAM Review* **51**(4), 661–703 (2009).
- [6] E. Herbert, G. Giraud, A. Louis-Napoléon and C. Goupil, “Macroeconomic Dynamics in a Finite World: the Thermodynamic Potential Approach”, [arXiv:2204.02038v2](#) (2022).
- [7] R. Wildauer and I. Heck, “Was Pareto right? Do the US wealth and income distributions follow power laws?”, Greenwich Papers in Political Economy, GPERC92, September 2024 version (series coverage: 2023), [institutional archive](#).
- [8] I. Wright, “The duration of recessions follows an exponential not a power law”, *Physica A* **345**, 608–610 (2005), [author’s text](#).
- [9] A. Aladangady and A. Forde, “Wealth Inequality and the Racial Wealth Gap”, *FEDS Notes*, October 22, 2021, [Federal Reserve](#).
- [10] Congressional Budget Office, *The Distribution of Household Income, 2019*, November 2022, [report and definitions](#).
- [11] Q. H. Vuong, “Likelihood Ratio Tests for Model Selection and Non-Nested Hypotheses”, *Econometrica* **57**(2), 307–333 (1989), [doi:10.2307/1912557](#).

Reproducibility. Conventions, point data and sources are given in the appendices. The author’s comments and the previous version are kept separately, and are not required to compile this article.

A Reading glossary

Entries follow the order of first appearance in the body of the text, after the abstract. Symbols introduced together are grouped; their operative definitions remain those of the corresponding sections.

Term or notation	Meaning in this article
Rebound effect	Deviation from the consumption savings expected after an efficiency gain. Here, the experiment measures a non-proportional productive response, without calibrating a final consumption.
Endogenous; emergence	Produced by the interactions and evolution of the system, rather than imposed as a macroscopic target.
Self-organized criticality (SOC)	Hypothesis of a regime spontaneously maintaining itself near a critical threshold. It is not demonstrated here.
Entity	Generic producing agent, able to lend, borrow and disappear; none of these roles is predefined.
K_i ; productive stock	Mobilizable physical capital of entity i , in model joules; distinct from its net worth.
Technology $f_i(K) = A_i K^{\gamma_i}$	Production added to the stock during a step; A_i is the extraction coefficient, γ_i the production exponent.
K°, γ°	Birth endowment and production exponent at birth in the homogeneous reference regime.
Principal q ; service rq	Nominal amount of a loan and interest due per step; the principal is not amortized.
C_i, D_i, NW_i	Claims, debts and net worth: $NW_i = K_i + C_i - D_i$.
$\lambda, \delta, \sigma, \rho$	Birth intensity, fraction depreciated per step, amplitude of the log-normal shock and intensity of market encounters.
Credit institution	Rules fixing the direction, amount and rate of loans; it does not here designate a particular bank.
Invariant; real balance sheet	Relation imposed by the model's operations. The real balance sheet tracks contributions, production and stock losses.
B_t, η_i	Number of births at step t and individual logarithmic shock, renewed at each step.
Concavity	Decrease of the marginal return with the stock, for $0 < \gamma < 1$; origin of the possible reallocation gain.
Service default	Inability to fully pay the interest due at the step considered.
Stock; flow; Δt	Quantity held; quantity transferred or produced during a step; possible physical duration of that step.
Exergy	Capacity of a resource to supply work relative to a reference environment; its correspondence with economic entitlements remains a hypothesis here.
K_{aut}, x_i	Deterministic fixed stock of an isolated entity with no noise or credit; reduced capital $x_i = K_i / K_{\text{aut}}$.
κ°, a	Reduced endowment K° / K_{aut} ; alternative coefficient $a = A(K^\circ)^{\gamma-1}$.
Rescaling; compensation	The former transforms the whole state and its units; the latter here adapts the endowment of future entrants to preserve κ° after the rise in A .
Tension T ; K_{eq}	Ratio $K_{\text{aut}} / K_{\text{eq}}$; fictitious stock giving the average production per producer. In the protocols, T can also denote the horizon of a simulation.
Y, N_{prod}	Total production of the step and number of entities present in the productive phase.
Marginal return $f'(K)$	Additional production at the margin per additional unit of stock.
L, Δ, p	Productive loss of the lender, joint productive surplus and share of that surplus attributed to the lender: $rq = L + p\Delta$.
Bankruptcy; cascade; avalanche	Removal of an entity in default or insolvent; propagation of losses; weakly connected component of losses among entities that died in the same step.
Root; generation; b_1, b_2	Initial death; rank of intra-step propagation; fraction of non-root deaths and average number of causal links between successive generations per death.
Seed; pairing; 95% CI	Initialization of the random generator; branches originating from the same state; 95 % confidence interval computed across repetitions, unless stated otherwise.
k ; stationarity	Historical size of the encounter pool, fixed at two in the main protocol; temporal invariance of statistical properties, not demonstrated by a single drift test.
Little's law	Relation between mean population, arrival flow and mean lifetime under conditions of stability and boundary control.
S, α, S_c, Z	Avalanche size, exponent, exponential cutoff and normalization constant of its fitted law.
Bootstrap; R^2	Resampling of whole realizations; descriptive goodness-of-fit score, not proof of a family of laws.
Contraction; D, β	Sequence of decreasing steps of a quantity; duration of that sequence and decay rate of its exponential survival. D without subscript in this context is not a debt.
Gini; Lorenz	Synthetic inequality index; curve of the cumulative share of a quantity as a function of the share of entities ranked by it.

Term or notation	Meaning in this article
Tail; Pareto; log-normal	Domain of large values; power-law decaying family; family whose logarithm is normal.
Finite elasticity ε	Ratio of the logarithm of the productive response to the logarithm of the technical factor m , here $m = 1.5$.
Quantile; decile; cohort	Ranking threshold; group of 10 % reranked at each date; set of individuals tracked by identity.
Covariance	Measure of joint variation used in the sharing weighting identity; it does not demonstrate causality.
Load ratio	Interest paid or due, depending on the definition, relative to production plus interest received; distinct from a debt-stock-to-production ratio.
Vuong	Comparison of model likelihoods on the same sample; its assumptions and limitations are made explicit in the appendix.

B Case study: one concrete realization

The previous sections aggregate observations over twelve seeds. This appendix instead details a single realization of the main control (seed 0, $T = 4000$, 60151 deaths, 8174 avalanches), to give a concrete face to the mechanisms discussed above. None of the figures that follow is a new statistical measurement: they illustrate, they do not replace the aggregated results in the body of the article.

B.1 Full trajectory

Figure 17 reprises the entire 4000 steps of this realization, without the restricted window of Figure 4. The startup phase, where the population grows from a single entity, stands out clearly from the fluctuating regime that follows; the article’s main analysis window ($3000 < t \leq 4000$, Appendix C) is marked on the plot.

B.2 Individual outcomes

Five entities from this same realization illustrate distinct outcomes: two reach $t = 4000$ (one, present from startup, at the top of the stock ranking; the other, born much later, at a median level); a third is the root of an isolated small avalanche (size 12, depth 9); a fourth is one of the thirteen roots of the largest avalanche of the stationary window (size 145, depth 20, detailed in Figure 20); the fifth is the most distant victim of that same avalanche, at generation 20.

Are these three deaths early or late? Figure 19 places their age within the full distribution of this seed’s 60151 lifetimes: the three are 8, 31 and 36 steps, against a median age of 13 steps over all deaths; none is an extreme longevity event, the tail of the distribution extending to 3916 steps.

B.3 Two cascades, two scales

Figure 20 explicitly draws the propagation graph of two real avalanches from this realization, in the same spirit as the stylized example of Figure 3 but from simulated data: a small avalanche (size 12, a single root, depth 9) and the largest of the stationary window (size 145, thirteen roots, depth 20). Both are drawn from the same seed; their difference in size is not a seed effect, but the variability already quantified in Figure 5.

These two sizes are not chosen to be representative: Figure 21 places them within the ranking of the 4086 avalanches of this same seed and window. Size 145 is the largest of the whole window (rank 1); size 12 is common, at rank 586, in the upper third of the distribution.

B.4 Chronology of the large cascades

Are large avalanches clustered in time, or dispersed? Figure 22 places every avalanche of this seed and window by its time and size. No visible temporal clustering of large sizes appears: the short contractions of aggregate activity (§4.3) therefore do not coincide with a

discernible agenda of individual large cascades; the two phenomena, though both defined on the same loss graph, are not the same observation.

B.5 Credit network structure at a given instant

The previous figures draw the loss graph, i.e. the propagation of defaults; this is not the credit network itself. The book of loans actually active at $t = 4000$ allows the distinction: each entity there has an out-degree (number of active claims, as a lender) and an in-degree (number of active debts, as a borrower), independent of any bankruptcy. Of the 1031 entities alive at that instant, 1023 carry at least one active loan; total degree (claims plus debts) is 54 at the median.

Figure 23 compares this distribution to that of two other seeds at the same instant: the shape is stable from one seed to another, so this is not a peculiarity of seed 0 chosen for this case study. The claims-versus-debts scatter also shows that few entities are strongly net creditors: the bulk of points sits near the diagonal or below it, and the few entities with the strongest active claims (above 80) do not symmetrically carry as much debt.

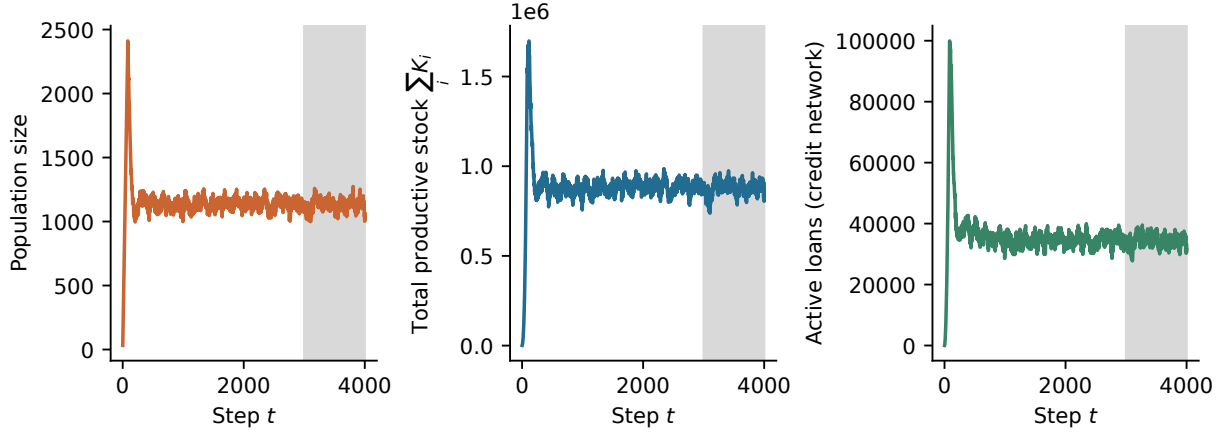


Figure 17: The entire realization, unaggregated. Main control, seed 0, $0 \leq t \leq 4000$. Left: population size. Center: total productive stock $\sum_i K_i$. Right: number of active loans, i.e. the size of the credit network itself. The three series share the same startup-then-fluctuating shape; the shaded zone marks the stationary window used elsewhere in the article.

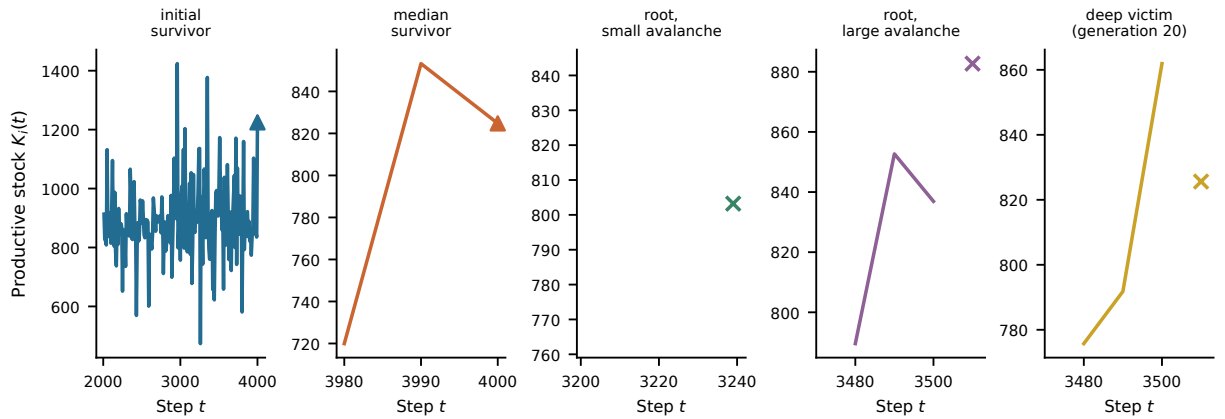


Figure 18: Five individual trajectories. Productive stock $K_i(t)$, time steps recorded every 10 steps. Each curve stops at the entity's death (marker, cause given in the exported data) or continues to $t = 4000$ for the two survivors. One entity may grow then disappear with no link to a large cascade; another may be swept away after several generations of propagation.

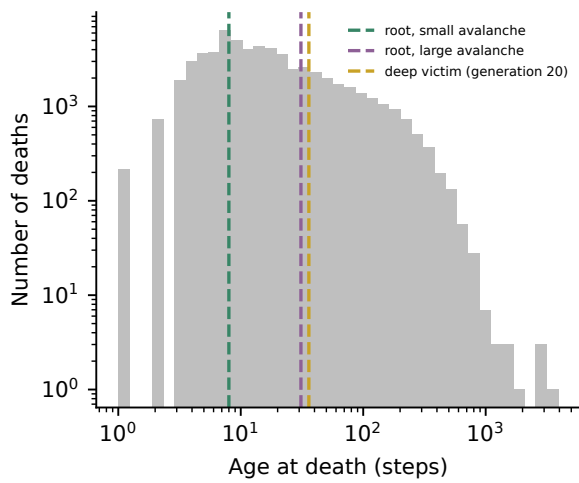


Figure 19: Where do these three entities sit in the lifetime distribution? Main control, seed 0, 60151 deaths over $0 \leq t \leq 4000$, median age 13 steps. Vertical lines: age of the three dead entities from the previous figure.

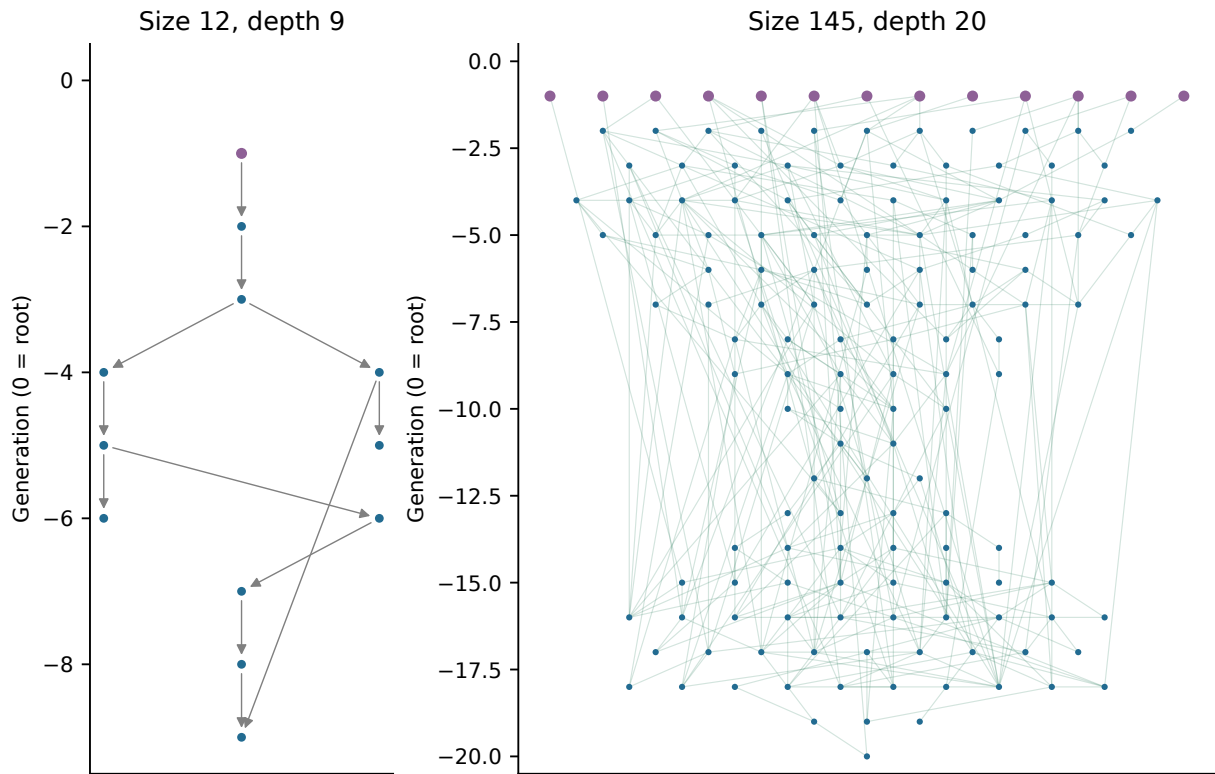


Figure 20: Two really simulated avalanches. Main control, seed 0. Left: avalanche of size 12, $t = 3239$ (12 nodes, 12 edges). Right: avalanche of size 145, $t = 3510$ (145 nodes, 313 edges), the largest of the stationary window for this seed. Each node is a dead entity; each arrow (or, on the right, each thin segment, for lack of room for arrowheads) goes from the defaulting borrower to the affected creditor (same convention as Figure 3), height codes the propagation generation. The number of edges far exceeds the number of nodes: a creditor may be affected by several deaths, this is not a tree in the strict sense, unlike the stylized example of Figure 3. Roots in solid color.

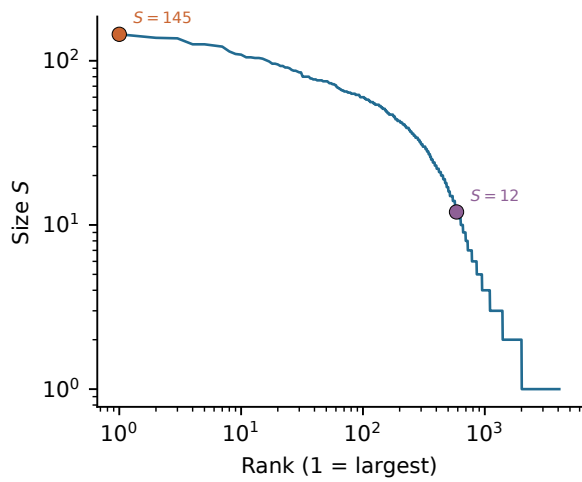


Figure 21: Rank of the two chosen avalanches within their own seed. Main control, seed 0, 4086 avalanches, $3000 < t \leq 4000$. The curve is the ranked size distribution (rank 1 = the largest); the two points mark the avalanches of Figure 20.

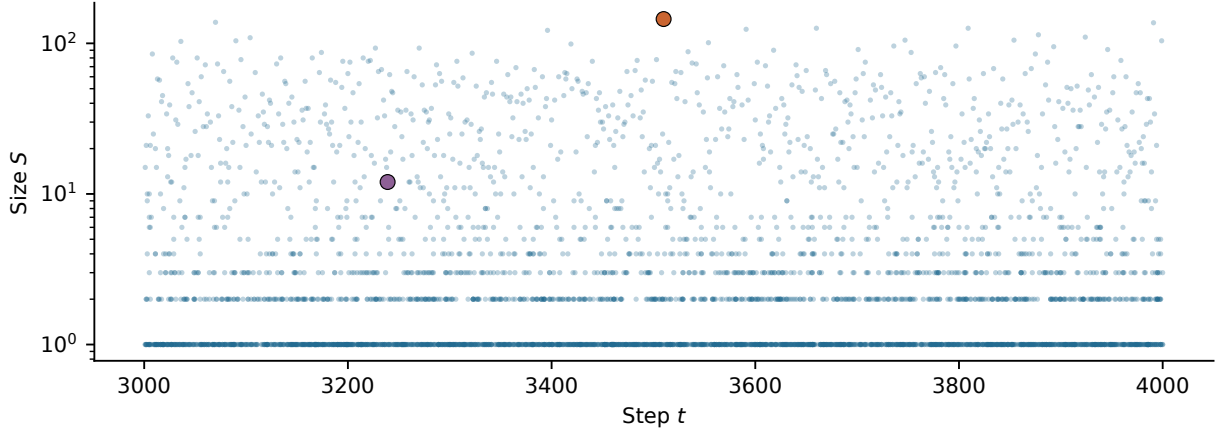


Figure 22: Size of each avalanche by its time. Main control, seed 0, 4086 avalanches, $3000 < t \leq 4000$. The two colored points reprise the avalanches of Figures 20 and 21.

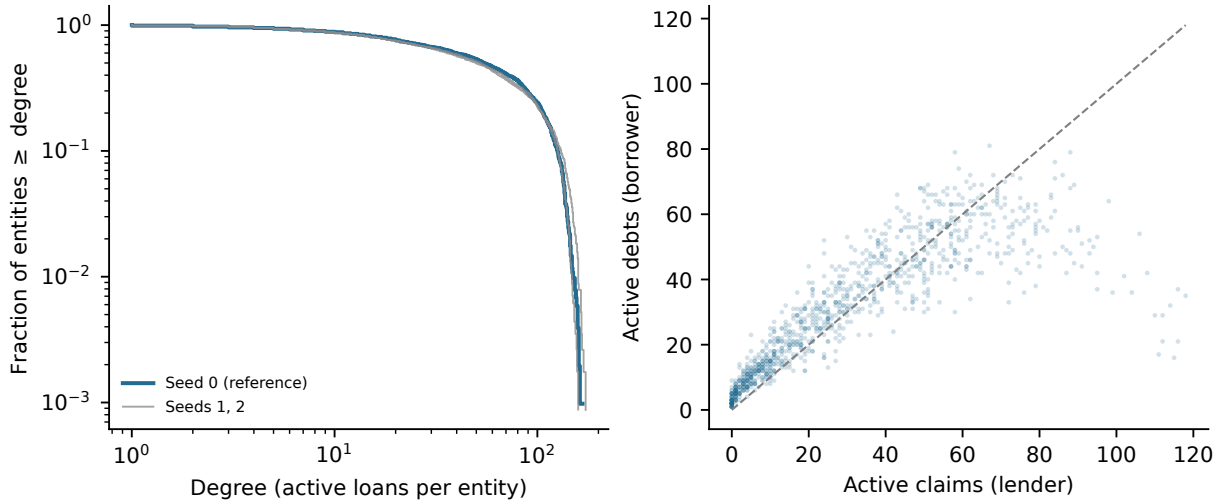


Figure 23: The book of active loans, not the loss graph. Main control, $t = 4000$. Left: distribution of total degree (claims plus active debts), seed 0 against two reference seeds (1 and 2); the shape is stable, not specific to one seed. Right: active claims against active debts per entity, seed 0; the diagonal marks lender/borrower equality.

C Checking the regime studied and the response calculations

For each seed s , arm and observable X , we compute

$$R_s(X) = \frac{\langle X \rangle_{3750 < t \leq 4000}}{\langle X \rangle_{3000 < t \leq 3250}}, \quad t_X = \frac{\bar{R}(X) - 1}{s_R / \sqrt{12}},$$

where s_R is the standard deviation of the twelve ratios. The 95% CI is $\bar{R} \pm t_{0.975;11} s_R / \sqrt{12}$, with $t_{0.975;11} \simeq 2.201$. This diagnostic compares two distant temporal levels; it analyzes neither the full correlation structure nor the stability of the distribution.

None of these contrasts exceeds the usual two-sided threshold, but the intervals allow drifts on the order of a percent. The temporal observations of a single run are not used as 1000 independent repetitions. The result does not automatically hold for an earlier window, another observable or an unrepresented exploratory configuration.

Renewal and memory of the initial state. The renewal of a decile provides another time scale, but the ranking quantity must be specified and the disappearance of an entity distinguished from its exit from the decile. For a set $D(t_0)$ defined at a reference date, retention $|D(t_0) \cap D(t_0 + h)| / |D(t_0)|$ measures persistence in the group, while the fraction of its members still alive measures their survival. These two curves are not interchangeable; the term "first decile" must specify whether it denotes low or high values. Even low retention does not demonstrate forgetting of the initial configuration: commitments and correlations may persist beyond individuals. This check must be confronted with the autocorrelations of the observables and with distinct initializations. No quantified forgetting time is established here; the distance from the final window and the absence of detected drift are not enough to certify it.

Population balances give exactly $N(t+1) - N(t) = B(t) - D(t)$ if B and D denote the entries and exits of the step under the same counting convention. Over a long stable window, $\bar{B} \simeq \bar{D}$ and $\bar{B} \simeq \lambda$; the ratio $\bar{D}/\bar{N}$ becomes close to $\lambda/\bar{N}$. This is not exactly the mean of $D(t)/N(t)$, and boundary terms are not zero by decree. The relation explains why a mortality elasticity can be almost the opposite of a population elasticity without constituting an independent causal finding.

The elasticity of equation (5) concerns a finite intervention. Its mean and CI are computed after forming the twelve paired log-ratios, not after averaging all time points in both arms. The decomposition formulas are sensitive to the choice of population: entities present at production may disappear before the end-of-step snapshot. This difference is preserved in the source data; it must not be erased to force the numerical closure of an ill-defined identity.

D Distribution fits: definitions and limitations

D.1 Mass, per-bin density and survival function

For an integer size S , the mass $P(S = s)$ and the survival $P(S \geq s)$ are distinct. The histogram here represents the mass of a bin $[l, r[$ divided by $r - l$, so that a wide bin does not appear more probable merely because it contains more sizes. The fitted curve is summed over the same integer sizes before this division. A survival slope cannot be substituted directly for the mass exponent.

The descriptive likelihood of the law (4) is maximized per seed, over all sizes $s \geq 1$. The numerical normalization uses $1 \leq s \leq 20000$; the mass of the last thousand values is checked to be below 10^{-10} at the optimum. The numerical search bounds are $-2 \leq \alpha \leq 5$ and $10^{-5} \leq 1/S_c \leq 2$; an optimum at a bound would signal insufficient identification. The 24 fits presented are inferior. The fitting support is not chosen to maximize the apparent straightness of a subset of points.

The cross-seed CIs and the bootstrap over whole runs measure the variability of the estimators or of the average curves obtained by this protocol. Fitting by product of probabilities does not demonstrate the independence of a run's avalanches. We therefore do not derive event-level standard errors or a certified goodness-of-fit test from it. In particular, this should be complemented by a comparison of families on the same support and an absolute-adequacy diagnostic accounting for temporal dependence [5].

The descriptive coefficient shown is

$$R_{\log}^2 = 1 - \frac{\sum_j [\log \hat{p}_j - \log p_{j,\text{fit}}]^2}{\sum_j [\log \hat{p}_j - \overline{\log \hat{p}}]^2},$$

over the non-empty empirical bins. This is not a percentage of correctly predicted probability, and it is not the criterion optimized by the likelihood. It depends on the binning. Zeros are not replaced by an arbitrary small value to make them loggable.

D.2 Durations: discrete exponential and window choice

If $D \geq 1$ follows a geometric law, with stopping probability u at each step, then $P(D \geq d) = (1 - u)^{d-1}$ and $\hat{u} = 1/\bar{D}$. Thus $\hat{\beta} = -\log(1 - 1/\bar{D})$ in $P(D \geq d) = \exp[-\beta(d - 1)]$. The regressions of figure 7 use this normalized form, not a log-log line. The R^2 there is computed on the non-zero log-survivals, at durations 1 to 11. Episodes remain potentially correlated and the window boundaries truncate some of them; the CIs are therefore based on the five seeds and not on a fictitious number of independent recessions.

D.3 The Vuong test and what it actually compares

In the existing analyses of net worth and interest, the threshold $x_{\min}$ is chosen by minimizing a Kolmogorov–

Table 2: Ratio of the means of the last and first 250 steps of the terminal window.

Arm	$R(Y)$ (95% CI)	$R(N)$ (95% CI)	$R(\sum K)$ (95% CI)
Control	1.0029 ± 0.0152	1.0014 ± 0.0132	1.0043 ± 0.0178
$A \times 1.5$	0.9968 ± 0.0139	0.9979 ± 0.0116	0.9957 ± 0.0167
$A \times 1.5$, compensated endowment	0.9987 ± 0.0132	0.9974 ± 0.0122	0.9998 ± 0.0148
Entrants $A \times 1.5$	1.0088 ± 0.0106	1.0070 ± 0.0090	1.0092 ± 0.0128
Entrants $A \times 0.75$	0.9906 ± 0.0159	0.9928 ± 0.0139	0.9905 ± 0.0184
Entrants $\gamma = 0.6$	1.0097 ± 0.0159	1.0038 ± 0.0133	1.0100 ± 0.0186

Smirnov distance for a continuous power law. Above this same threshold and on the same values, a log-normal and an exponential are also fitted, conditioned on $X \geq x_{\min}$. The power law compared here has *no upper cutoff*: this is not the discrete model with exponential cutoff used for avalanches. The *left* truncation, which defines a tail sample, must not be confused with the *right* cutoff.

For the fitted densities f and g , define $d_i = \log f(x_i) - \log g(x_i)$ and then

$$z = \frac{\sqrt{n} \bar{d}}{s_d}, \quad s_d^2 = \frac{1}{n-1} \sum_i (d_i - \bar{d})^2.$$

A positive value favors f in average log-likelihood. The usual interpretation of ± 1.96 as a 5% threshold requires regularity, model-separation and sampling assumptions. Here the threshold was selected, entities interact, and some fitted log-normals are very close to a power-law limit. The score is therefore reported as a diagnostic, not as certification of these assumptions [11].

The flag $\sigma_{LN} > 10$ concerns 50 of the 72 interest fits and 46 of the 72 net-worth fits. It marks a numerical zone, not a mathematical boundary between two families. Two visible groups in a scatter of fit parameters do not mean that the agents form two economic populations: each point in that scatter represented one fit, not one entity. Per-fit results and their identifiers are available in `data_revision/rev_vuong_points.csv`.

Comparing Pareto, cutoff power law and log-normal requires fixing the same domain, describing the threshold selection and evaluating goodness of fit, not merely counting parameters. A cutoff that diverged with system size would define an interesting limit, to be checked across several sizes; it does not make its finite-size fit free of charge.

E Credit load and surplus weighting

E.1 Definition and measurement of the load ratio

Let P_i denote the step's production, I_i^{received} the interest received, I_i^{paid} that actually paid, and D_i the sum of

principals due. Three ratios must be distinguished:

$$c_i^{\text{paid}} = \frac{I_i^{\text{paid}}}{P_i + I_i^{\text{received}}},$$

$$c_i^{\text{due}} = \frac{\sum_{\ell:i \text{ borrows } \ell} r_{\ell} q_{\ell}}{P_i + I_i^{\text{received}}},$$

$$d_i^{\text{income}} = \frac{D_i}{P_i + I_i^{\text{received}}}.$$

The first is a share of gross income paid; the second is the contractual charge due; the third relates a debt stock to a per-step income and is interpreted in a number of steps. The variable D_i/P_i represents neither of the first two ratios. The denominator includes financial income, even though its aggregate sum is not new real production.

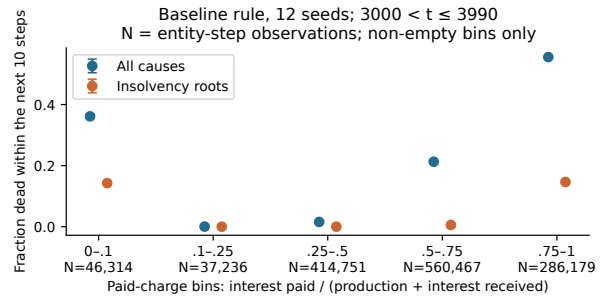


Figure 24: A correctly named load diagnostic. Marginal rule M4.4, twelve seeds. For the snapshots $3000 < t \leq 3990$, the variable is $I_i^{\text{paid}}/(P_i + I_i^{\text{received}})$; the response is the fraction of entities dying in the following ten steps. The bins have the explicit bounds shown, common to all seeds; they are not capital classes. The last incomplete window is excluded. The 95% CIs compare the fractions obtained across twelve runs, without attributing binomial independence to repeated entities. All-cause deaths and insolvency roots are shown separately.

The panels record the interest paid, not the service due at the exact moment preceding payment. A pro-rata payment can make an unsustainable load look low; the moment of collection and the flows received within the step also matter. The diagram remains descriptive, conditional on the survivors observed in the panel. It justifies neither a causal link from load to death nor a general refutation of convexity. Convexity calculations or Jensen-gap computations on D_i/P_i would not answer the question asked here about the load ratio as defined.

E.2 Proof of the weighting identity

For n contracts whose mean surplus is strictly positive, let $\bar{p} = n^{-1} \sum p_j$, $\bar{\Delta} = n^{-1} \sum \Delta_j$ and $p_w = \sum p_j \Delta_j / \sum \Delta_j$. Defining the empirical covariance with denominator n ,

$$\text{Cov}_n(p, \Delta) = \frac{1}{n} \sum_j (p_j - \bar{p})(\Delta_j - \bar{\Delta}) = \frac{1}{n} \sum_j p_j \Delta_j - \bar{p} \bar{\Delta},$$

we immediately obtain

$$p_w - \bar{p} = \frac{\text{Cov}_n(p, \Delta)}{\bar{\Delta}}. \quad (6)$$

If a corrected covariance with denominator $n - 1$ is used, the numerator must be multiplied by $(n - 1)/n$. The identity also holds in expectation when the moments exist and $\mathbb{E}[\Delta] \neq 0$.

For a constant fixed sharing rule, the gap is zero by construction. For a variable rule, it can be positive even if most contracts have a very small principal or surplus. The effective distribution of contracts determines its magnitude; an analytical grid covering rarely realized contracts cannot replace this distribution. Qualifying a rule from these theoretical cases alone would overstate their empirical reach. The accounting identity does not demonstrate an explanation of mortality or population differences between institutions.

F Measurement supplements and tabulated values

F.1 Quantile ratios

The table below and figure 14 rest on the same observations. Each $\pm$ is a half 95% Student CI of the twelve per-seed ratios, not a standard deviation across entities. "Gross income" means production plus interest received.

Quantity	q10	q50	q99	q99.9
Production	1.748 ± 0.004	1.806 ± 0.003	1.831 ± 0.004	1.816 ± 0.008
Gross income	1.755 ± 0.004	1.776 ± 0.007	1.741 ± 0.070	1.474 ± 0.060
Net worth	1.103 ± 0.007	1.486 ± 0.016	1.452 ± 0.095	1.169 ± 0.065
Capital	1.386 ± 0.008	1.451 ± 0.005	1.489 ± 0.006	1.460 ± 0.013
Interest received	0.111 ± 0.025	1.743 ± 0.013	1.726 ± 0.081	1.447 ± 0.062

F.2 Temporary reach of an intervention

An intervention limited to the entities present can die out with their cohort if entrants do not inherit it. This partial-reach experiment is a protocol control, not a main proof of the rebound mechanism. Conversely, modifying all entities and future entrants preserves the treatment despite population renewal. Modifying γ changes the shape of the function and the dimensions of its coefficient: this is not the same operation as multiplying A at fixed γ .

F.3 Multiple losses and sufficient causes

The graph can attribute several losses to a victim. This does not mean that each would have sufficed, in isolation, to make it insolvent. To illustrate this, an initial net worth of 5 and two losses of 3 lead to -1 ; no single loss suffices. With a net worth of 2, each of the two losses would suffice. It is the solvency margin before the losses that makes the difference, not merely the number of edges.

The count of parents must also specify whether it retains only losses coming from the previous generation or all those of the step. A detailed sufficiency analysis also requires reconstructing the debts written off; it is not necessary to establish the existence of propagation nor the non-proportionality of the response. Earlier calculations remain archived, but their percentages are not used as evidence in this article.

F.4 Stock and production on the same realization

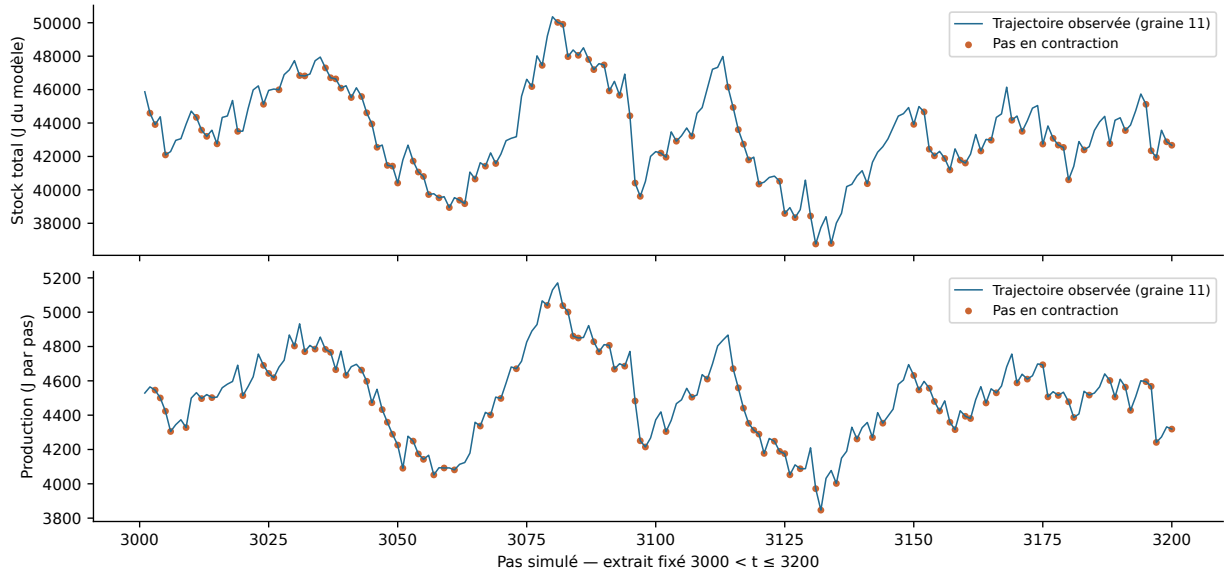


Figure 25: Observing the difference between stock and flow. M4B central, seed 11, excerpt $3000 < t \leq 3200$. Each line follows a single realization, with no measurement uncertainty artificially added; the orange points mark the down-steps. The contraction durations are computed separately on the two observables in figure 7.

G Local sources, configurations and reproducibility

The PDF can be compiled with only the LaTeX files and figures in this folder, without the internship report or the questionnaires. The new calculations are produced by `../scripts/revision_article.py`, which writes only into this article's folder and launches no simulation. The numerical exports, per-seed observations and SHA-256 checksums of the sources are in `data_article/`. The retained supplements from the previous version are in `data_revision/`. Recomputing from the microdata requires the local campaigns, which are not all version-controlled.

G.1 Three protocols, samples not merged

Family	Configuration and data
M4B central	$A = 1$, $\gamma = 1/2$, $\lambda = 30$, $\delta = 0.05$, $\sigma = 0.25$, $K^\circ = 25$, $k = 3$, geometric target. Five seeds 11–15; contractions on $1000 < t \leq 4000$, Gini on the final state. Simulation Lab identifiers are in the episode and confirmation exports. The Gini sweep shown varies σ only.
Live-v1 recalibration	Reference $A = 1$, $\gamma = 1/2$, $\lambda = 30$, $\delta = \sigma = 0.01$, $K^\circ = 25$, $\rho = 1$; seeds 0–2, empty start. Window $1200 < t_{\text{old}} \leq 2000$. The fine-depreciation arm uses its own reference $\delta = 0.002$. The folders are named in <code>temps_graines.csv</code> .
M4.4 main	Same physical reference as Live-v1; arithmetic target in the homogeneous regime, free loan direction, marginal rule. Twelve seeds 0–11, intervention at step 2001, window $3000 < t \leq 4000$. Compensated endowment: $K^\circ = 56.25$ for new entrants after the intervention; existing stocks unchanged.
M4.4 sharing	$p \in \{0; 0.25; 0.5; 0.75; 1\}$ or marginal rule; twelve seeds, same window. Ratios to the $p = 0$ arm are paired. Load diagnostic: marginal rule only, $3000 < t \leq 3990$, ten-step death horizon.
M4.4 encounters	$\rho \in \{0.5; 1; 1.5; 2; 3\}$; twelve seeds per value. Balance-sheet Ginis are recomputed on the panels of $3000 < t \leq 4000$, like the windows of the other measurements.

G.2 Each figure and its data-point files

The CSV files with no prefix and `diagnostic_chaine.json` are relative to `data_article/`. The TeX files and paths starting with `data_revision/` are relative to the LaTeX folder. The manifest links the exports to the original sources and their checksums; it does not turn an aggregated export into complete microdata.

Fig.	Data / local file	Nature of the calculation
1	<code>exemple_article.tex</code>	Four fictitious entities, real loan and nominal recording; no simulated measurement.
2	<code>technologie_points.csv</code> , <code>technologie_courbes_v2.csv</code>	Analytical technologies $2K^{1/4}$ and $K^{1/2}$; no regression.
3	<code>cascade_article.tex</code>	Fictitious five-entity balance sheet; four-bankruptcy tree checked separately.
5	<code>avalanches_counts.csv</code> , <code>avalanches_fits.csv</code> , <code>avalanches_points.csv</code>	Integer sizes in the avalanche CSVs; fits and cross-seed CIs.
7	<code>cycles_fits.csv</code> , <code>cycles_points.csv</code> ; episodes in <code>data_revision/rev_cycles_episodes.csv</code>	Five M4B runs; exponential coefficient from the mean duration.
8	<code>lorenz_points.csv</code> , <code>lorenz_ginis.csv</code>	Control panels; exact dates 2010, 3000, 4000.
9	<code>m4b_sensibilite_points.csv</code> , <code>m4b_nw_confirmation.csv</code>	NW Gini on five independent final snapshots per cell.
10	<code>rho_points.csv</code> , <code>rho_fits.csv</code> , <code>rho_nw_graines.csv</code>	Three estimators from the campaign table, plus recomputed NW Gini; per-seed regressions.
11	<code>temps_graines.csv</code> , <code>temps_points.csv</code>	Live-v1 series rescaled to a common duration and a common flow per old step.
12	<code>elasticites_graines.csv</code> ; curves in <code>data_revision/rev_reponse_courbes.csv</code>	Elasticities recomputed from the series; centered 101-step smoothing kept for the curves.
13	<code>corps_article.tex</code> ; diagnostics <code>diagnostic_chaine.json</code>	Interpretive diagram, not a causal graph identified by intervention on each mediator.
14	<code>data_revision/rev_quantiles_graines.csv</code>	Per-seed ratios kept; abscissa transformed to logit.
15	<code>deciles_v2_points.csv</code> , <code>deciles_v2_graines.csv</code>	Shares of production, interest and NW; ranked by capital at each snapshot.
16	<code>institutions_points.csv</code> , <code>institutions_fits.csv</code> , <code>institutions_references.csv</code>	Ratios to $p = 0$ and descriptive linear regressions.
24	<code>charge_graines.csv</code> , <code>charge_points.csv</code> , <code>charge_resume.csv</code>	Interest paid / gross income; deaths observable throughout the horizon.
25	<code>data_revision/rev_stock_trajetoire_points.csv</code>	Raw M4B excerpt; figure kept.

The new age-composition checks are in `ages_graines.csv`. The ranking conventions are explicit in the script; ties are broken by stable order. The revision plan and the decision register are outside the LaTeX folder, in `../revision_2026-09-15/`, to keep the corrections to be reported in the internship report separate.

The supplement on instantaneous median ages is computed by `../scripts/ages_v2.py`. Its per-seed values, its averaging convention and the checksums of the source panels are kept in `../revision_2026-09-17/ages_v2_graines.csv` and `../revision_2026-09-17/ages_v2.json`. It runs no simulation and modifies no figure generator.

The detailed configurations and the M4.4 identifiers of the initial campaign remain in `data_revision/runs_m44.csv`, which distinguishes the recorded parameters from the interventions. Some seeding metadata does not record the parameters; these gaps are not filled by silent deduction. The technical K0 codes from the source files are kept for their traceability; in the scientific notation, the endowment is now read K° and its reduced ratio κ° . Renaming the notation does not modify the engines.

G.3 Engine compatibility and rule changes

Three levels of control must be distinguished: preserving the functionalities of the simulation lab, reproducing a trajectory in a reference regime, and comparing two models whose rules differ. Only the second bears directly on dynamic parity.

The M4.2 report, section "M4B Parity", documents tests at $A = 1$, $\gamma = 1/2$ and $k = 2$: identity of discrete variables and comparison of floats at relative tolerance 10^{-9} . The reported 1000-step probe shows no discrete divergence, with a maximum relative deviation of the series of $3.1 \cdot 10^{-12}$. This finding is not a bit-for-bit identity nor a guarantee at every horizon.

For M4.4, in the campaign folder `m4_4_rebond_credit_soc/`, the file `results/analysis/parity_deviations_8000.csv` holds 8000 maximum deviations, all zero, over the 26 columns compared to the reference M4.3 run, in the homogeneous regime with arithmetic target. The test `tests/test_parity_m4_3.py` defines this scope. The test `tests/test_v1_equivalence.py` separately compares Live-v1 to the compatibility mode `richest_lends`, with a heterogeneous intervention; a negative control checks that freeing the loan direction can change the trajectory. The presence of this test defines a verifiable requirement; it does not by itself replace an execution report for every version.

The M4.4 campaign paths are kept as provenance identifiers. Switching to a different market rule must be distinguished from an instrumentation extension, without renaming the archives the data come from. Non-regression therefore does not allow merging the M4B, Live and M4.4 samples.

G.4 Why temporal aggregation is not a symmetry

A counterexample appears even before introducing the market. With no noise, credit or births, let $F(K) = (1 - \delta)(K + AK^\gamma)$. Two successive steps give

$$F(F(K)) = (1 - \delta)^2(K + AK^\gamma) + (1 - \delta)A[(1 - \delta)(K + AK^\gamma)]^\gamma.$$

An aggregated step, with $A' = 2A$ and $1 - \delta' = (1 - \delta)^2$, would instead give

$$G(K) = (1 - \delta)^2(K + 2AK^\gamma).$$

In general $F \circ F \neq G$: the second production is computed on an already modified stock. For example, at $K = 100$, $A = 1$, $\gamma = 1/2$ and $\delta = 0.01$, the values are approximately 118.142 and 117.612. The market adds

an alternation of encounters, contracts, interest and bankruptcies; composing the Poisson laws or the shocks alone does not compose these coupled operations.

The Live-v1 trials of figure 11 measure the collective reach of this difference. After full conversion of the parameters, aggregating two steps gives, relative to its reference, $N : 1.067 \pm 0.018$, $K_{\text{tot}} : 1.158 \pm 0.030$ and $Y : 1.118 \pm 0.024$. Aggregating four steps, the production ratio is 1.346 ± 0.026 . The protocol starts from an empty state, uses three seeds and compares $1200 < t_{\text{old}} \leq 2000$; flows are divided by the aggregation factor, not the stocks. The per-seed values and other variants are in `data_article/temps_graines.csv` and `data_article/temps_points.csv`. These trials and the counterexample establish the absence of an exact symmetry; they do not constitute a study of convergence toward a continuous limit.